ARTICLES

Publications most relevant to the proposal

* Corresponding author † Equal contribution.

1. S. Manti* and A. Scordo, “Recent Developments of the VOXES Von Hamos X-Ray Spectrometer for Laboratory XES and XAS Studies,” Engineering Innovations 19, 23–29 (2026). doi:10.4028/p-rQpe9Z

7 pages · combined pages 2–8

2. S. Manti*, F. Napolitano, A. Clozza, C. Curceanu, G. Moskal, K. Piscicchia, D. Sirghi and A. Scordo, “Enhancing Performances of the VOXES Bragg Spectrometer for XES Investigations,” Condensed Matter 9, 19 (2024). doi:10.3390/condmat9010019

9 pages · combined pages 9–17

3. S. Manti*†, S. H. Yip† et al., “Machine-learning-enhanced event selection for BEGe detectors in the VIP experiment,” Journal of Instrumentation 21, P01036 (2026). doi:10.1088/1748-0221/21/01/P01036

13 pages · combined pages 18–30

4. S. Manti*, F. Sgaramella, L. Abbene et al. (SIDDHARTA-2 Collaboration), “Precision test of bound-state QED at intermediate Z with kaonic neon,” Physical Review A 113, 022815 (2026). doi:10.1103/rry5-tdqb

8 pages · combined pages 31–38

5. K. Piscicchia, S. Donadi, S. Manti*, A. Bassi, M. Derakhshani, L. Diósi and C. Curceanu, “X-Ray Emission from Atomic Systems Can Distinguish between Prevailing Dynamical Wave-Function Collapse Models,” Physical Review Letters 132, 250203 (2024). doi:10.1103/PhysRevLett.132.250203

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Recent Developments of the VOXES Von Hamos X-Ray Spectrometer for Laboratory XES and XAS Studies

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Keywords: VOXES; XES; XAS; Von-Hamos

Abstract. VOXES is a Von Hamos X-ray spectrometer developed at the INFN National Laboratories of Frascati for high-resolution laboratory X-ray spectroscopy in the 5–20 keV range. It uses curved mosaic crystals and motorized positioning stages to perform wavelength-dispersive X-ray fluorescence (WD-XRF) with sub-10 eV tunable resolution for extended and dilute samples. Recent developments include the integration of an energy-dispersive X-ray fluorescence (ED-XRF) line based on a silicon pin-diode detector, which enables flux monitoring and simultaneous ED and WD measurements. In addition, a dedicated liquid-sample holder has been introduced, and a Y-shaped support geometry, crucial for switching to a transmission layout, provides mechanical compatibility with laboratory XAS, now under implementation. These upgrades expand the versatility and automation of VOXES, strengthening its role as a table-top platform for laboratory X-ray spectroscopy.

Introduction

X-ray spectroscopy [1], in its various forms, offers a well-established method for probing the state of matter under different physical and chemical conditions. Different X-ray techniques offer specific capabilities and applications across diverse fields such as materials science [2], chemistry and catalysis [3] with applications for battery research [4, 5], gemmology [6], cultural heritage [7], environmental analysis [8], and industrial quality control [9].

Among the techniques that detect X-rays, core-hole spectroscopies [10], such as X-ray Absorption Spectroscopy (XAS) and X-ray Emission Spectroscopy (XES), play a central role in probing the local electronic environment of atoms. The creation of a deep-core vacancy and its subsequent relaxation encode detailed information about oxidation state, spin configuration, ligand environment, and bonding characteristics [11].

Traditionally, these measurements mostly have been performed at large-scale facilities, such as synchrotrons and X-ray free-electron lasers, which provide highly brilliant and tunable X-ray beams [12]. Despite their remarkable performance, access to these infrastructures remains limited due to high demand, restricted beamtime availability, and the logistical constraints associated with conducting experiments in a facility. This has driven a growing interest in developing laboratory-scale X-ray spectroscopy setups [13, 14] capable of delivering high-resolution measurements using compact and cost-effective X-ray sources like X-ray tubes [15]. For instance, several table-top setups have been demonstrated for XAS and XES measurements [16], including configurations capable of performing both simultaneously [17, 18].

In this scenario, the VOXES [19, 20] spectrometer at the INFN National Laboratories of Frascati represents an example of how optimized optical design, advanced detectors, and precise motorized positioning systems can significantly enhance the performance of laboratory XES for large sources. Originally developed for application of kaonic atom X-ray spectroscopy at DAΦNE [21] within the SIDDHARTA-2 experiment [22], VOXES has evolved into a versatile platform capable of measuring emission spectra from extended or dilute samples, with a tunable sub-10 eV energy resolution in the 5–20 keV range [23].

In this article, we present the recent developments of the VOXES spectrometer. We first describe the main features of the setup and its operation in emission geometry for extended sources. We then introduce the design upgrades implemented to improve sensitivity and versatility. Finally, we show

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how the system can be reconfigured between emission mode (XES) and transmission mode (XAS), enabling both types of measurements within the same compact platform. The paper is structured as follows. Section 2 describes the operating principle of the VOXES setup. Section 3 presents the updated configuration, which allows switching between XES and XAS modes. Section 4 concludes the paper.

VOXES Experimental Setup

The VOXES experimental setup is shown in Fig. 1. Compared to previous publications [23], the setup now consists of two complementary detection lines: a Wavelength-Dispersive (WD)-XRF line based on a Von Hamos geometry with a mosaic crystal, and an Energy-Dispersive (ED)-XRF line using a compact silicon detector. The entire spectrometer is mounted on a motorized rail system to ensure reproducible positioning and remote operation.

The X-ray source is a MicroXray WindChill tube with a molybdenum anode. The Mo Kα line at

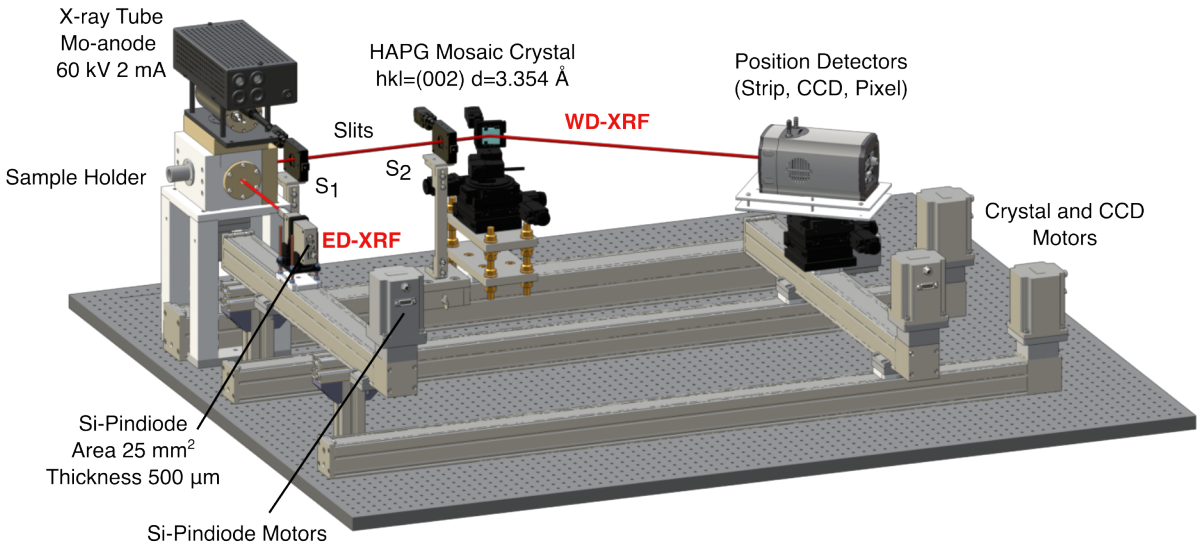


Fig. 1: The VOXES experimental setup. The X-ray tube on the left directs the beam onto the sample holder. After emission, the beam is conditioned by adjustable slits and motorized stages to ensure precise alignment. The resulting X-rays are diffracted by the mosaic crystal and subsequently detected by the position detector placed on the right.

17 keV results in a peak-free continuum in the 5–12 keV range, that is ideal for studying $3d$ transition metals without source-line contamination. The WD-XRF line consists of a slit system, a curved mosaic crystal, and a position-sensitive detector. Two motorized STANDA slits (S_1 and S_2) shape the beam emerging from the source box and define the effective source size and angular acceptance. Typical settings are an effective source width of $S_0' = 1$ mm and an angular acceptance $\Delta\theta' = 0.4°$ [20], allowing a controlled trade-off between spectral resolution and photon flux. The mosaic crystals used in VOXES are highly annealed pyrolytic graphite (HAPG) crystals, all oriented along the (002) direction with a lattice spacing of $d = 3.354$ Å. They share the typical properties of HAPG, namely high reflectivity, low absorption in the 5–20 keV range, and a controlled mosaic spread that enables efficient diffraction of divergent beams, with a declared mosaicity of $0.10° \pm 0.01°$. The specific crystals available for VOXES differ in their bending radius and thickness, allowing the instrument to be optimized for different applications. A set of cylindrical crystals is available with radii $\rho_c = 10.3$ mm,

77.5 mm, 103.4 mm, and 206.7 mm, as well as thicknesses of 20 μm, 40 μm, and 100 μm. The crystal mount is actuated by a five-axis STANDA positioning system, enabling fine control of its position and orientation with respect to the Bragg angle θ_B.

The diffracted spectrum is recorded with a position-sensitive detector. Several detectors are available depending on the target energy range and efficiency requirements. For one-dimensional detection, the setup can employ a MYTHEN2 strip detector (DECTRIS) with a 50 μm strip pitch and an active area of 32×8 mm^2. When two-dimensional imaging is required, particularly for applications where the vertical alignment of the dispersed spectrum is critical, an ANDOR iKon-M CCD camera can be used, featuring a 1024×1024 pixel sensor with a 13×13 μm pixel pitch and an active area of approximately 13.3×13.3 mm^2. Alternatively, a Timepix detector with a 2×2 module configuration is available, providing a total active area of 512×512 pixels with a 55 μm pixel size. All position-sensitive detectors are mounted on a motorized linear stage combined with a fine rotation axis, enabling precise positioning and optimal focusing of the Bragg-diffracted spectrum along the dispersion direction.

FESTO electrical stage motor and STANDA stepper motors provide full motorization of the slit sys-

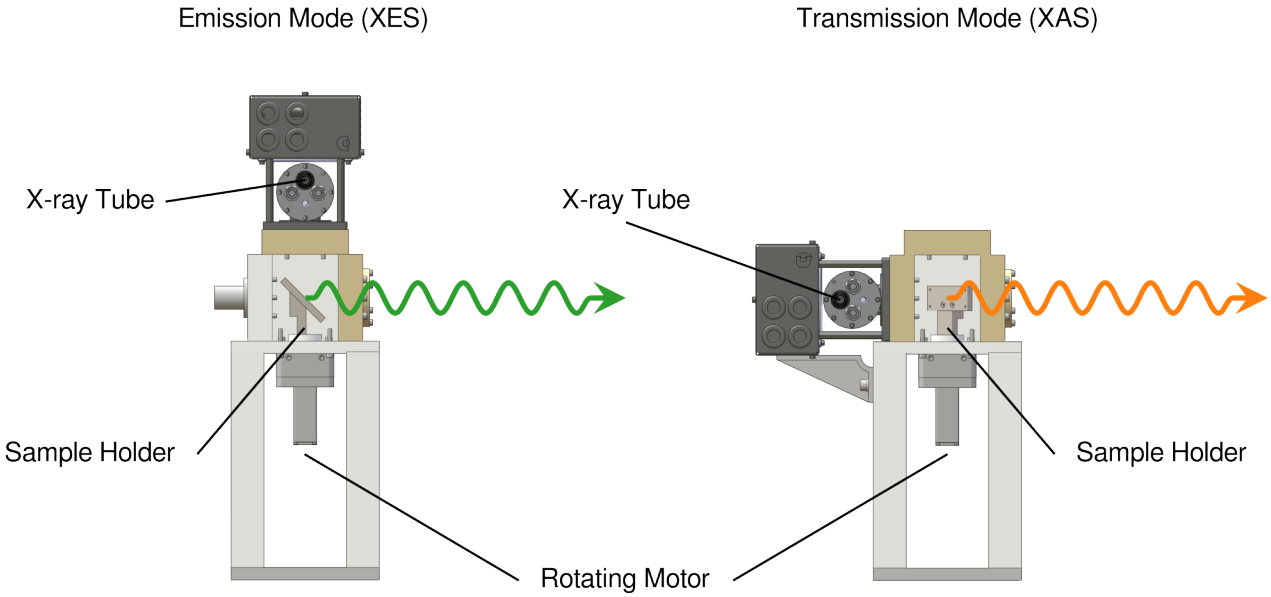


Fig. 2: VOXES spectrometer in transmission (left) and emission (right) modes. In transmission mode, the beam passes through the sample for XAS measurements. In emission mode, the emitted X-rays are detected for XES analysis. The setup allows switching between the two modes by rotating the source and detector assemblies.

tem, crystal mount, and detector stage. This ensures stable, repeatable alignment during scans and allows remote operation during long measurements. Fine adjustments of the Bragg angle and detector position are controlled through the motorized five-axis stages, which also provide pitch and roll corrections to maintain optimal focusing.

The ED-XRF line can operate in parallel with the WD-XRF system and employs an Amptek X-123 silicon pin-diode with integrated preamplifier and digital pulse processing. The detector uses a 25 mm^2 active area with approximately 500 μm silicon thickness and a thin Be window ($\approx$12.5 μm) for optimal transmission of low-energy X-rays. It delivers an energy resolution of 125–140 eV FWHM at 5.9 keV. The sample holder is mounted on a motorized rotation stage, allowing the sample to be oriented toward either the WD-XRF or ED-XRF line. It includes a Y-shaped aperture, which is used for optical alignment in emission geometry and also enables transmission of the X-ray beam when

Table 1: Position-sensitive detectors available in the VOXES spectrometer and their main characteristics and intended use cases.

Detector	Type	Advantages	Limitations	Applications
MYTHEN2	1D Strip	High count-rate and fast readout	No vertical information	Routine WD-XRF, calibration, high-statistics studies
ANDOR iKon-M	2D CCD	High spatial resolution, low noise	Limited count rate, slow readout	High-resolution XES, alignment and imaging
Timepix	2D Pixel	Single-photon sensitivity, flexible 2D imaging	Lower count rate	Low-flux measurements

the setup is reconfigured for XAS for future developments. The holder accepts custom mounts for different sample types; dedicated Teflon cells with thin Kapton windows were fabricated for liquid samples, enabling measurements of aqueous or organic solutions without realigning the spectrometer.

VOXES Operating Modes

The VOXES spectrometer can operate in multiple configurations depending on the measurement requirements. Its fully motorized geometry allows switching between WD-XRF and ED-XRF, with minimal manual intervention like shown in Fig. 2. The modular design also enables a transmission configuration that provides compatibility with XAS measurement, currently under development. In emission mode, the sample is positioned on the sample-holder at 45° so that it is directly irradiated by the X-ray tube. The emitted X-rays then impinge on the mosaic crystal, where their distance depends on the specific Bragg angle for the emission line under study. The mosaic crystal diffracts the photons, and the dispersed spectrum is recorded by the position-sensitive detector, one-dimensional with a strip detector or two-dimensional with a CCD or pixel detector. Motorized slits (S_1 and S_2) allow adjustment of the effective source size and angular acceptance, allowing a balance of energy resolution versus counting rate, depending on the target requirements. This geometry is used for high-resolution studies of emission lines and for laboratory WD-XRF measurements.

An example of the VOXES WD-XRF operating modes is shown in Fig. 3 obtained with the MYTHEN2 detector. The left panel displays a spectrum acquired from a FeCoNi metallic foil. This foil is used during alignment and energy calibration, taking advantage of the well-separated Co K$\alpha_{1,2}$ and Fe Kβ lines. Once the setup is aligned and calibrated, the same geometry can be used to investigate the electronic configuration of iron-containing samples. The right panel shows two preliminary XES measurements on iron compounds in different spin states: high-spin for $FeSO_4$ and low-spin for K_4FeCN_6. Spectra were acquired using the X-ray tube at 25 kV and 2 mA, with a total acquisition time of 4 hours. Due to the different spin configuration of iron, high and low respectively, the Kβ' satellite is strongly suppressed in the low-spin compound, while it is clearly visible in the high-spin case [24] and a few eV shift. This demonstrates the ability of the VOXES spectrometer to resolve chemical and spin-state dependent features in the Fe K$\beta_{1,3}$ emission line.

The ED-XRF line can operate independently and in parallel with the WD-XRF system. The silicon pin-diode detector is mounted on a motorized stage that allows it to be positioned closer to the sample to increase the detected signal, or retracted to avoid excessive dead time when the emitted flux is high. It collects energy-dispersive fluorescence spectra in the 2–25 keV range. This mode is typically used to monitor the emitted flux, particularly from dilute samples such as liquids, or as a complementary detection channel for quantitative analysis. The sample holder includes mounts for both solid and liquid samples (see Fig. 4). Liquid measurements are carried out using dedicated Teflon pouches with external dimensions of $25 \times 80 \times 2.5$ mm^3 and an exposed active area of 15×60 mm^2, encapsulated with 7 μm thin Kapton windows. This thickness results in a transmission of approximately 40% at 7

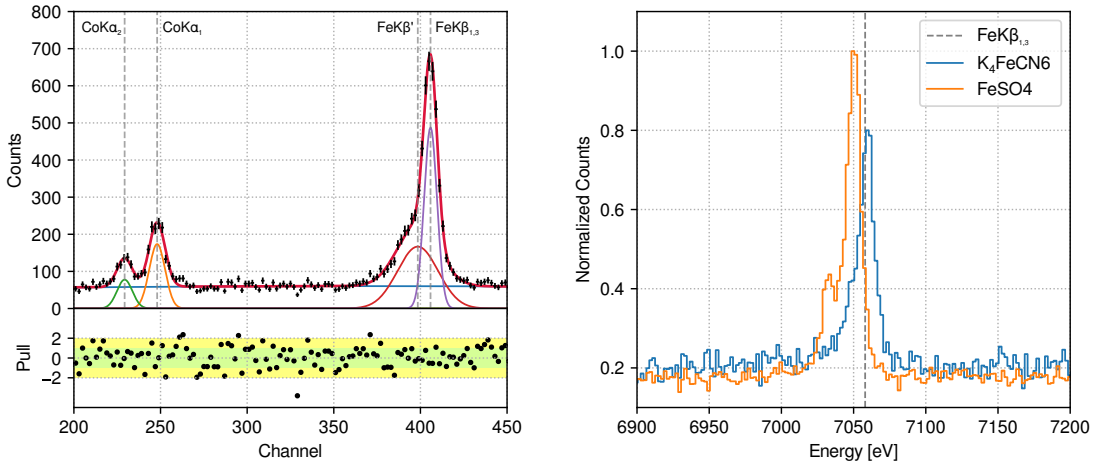


Fig. 3: WD-XRF Spectrum of a FeCoNi foil (left) used to align and calibrate with cobalt and iron reference line. Spectra (right) of high-spin K_4FeCN_6 (blue) and low-spin $FeSO_4$ (orange). The Fe Kβ' suppression in low-spin compounds is clearly resolved, indicating sufficient sensitivity for chemical-state analysis.

keV for water-like liquid solutions, which is close to the optimal regime for transmission-mode XAS measurements [13]. Other path lengths are available and can be selected depending on the photon energy and sample composition. The cells can be installed without modifying the optical alignment, allowing the system to switch rapidly between solid and liquid samples while preserving the same measurement geometry. The VOXES setup has been mechanically upgraded to allow operation in a transmission geometry suitable for laboratory XAS. This configuration is enabled by a Y-shaped sample holder and a reversible X-ray tube mount, which allows the beam to illuminate the sample from behind. Although XAS measurements are still under development, the hardware required to perform absorption studies is already in place and will be validated in future work.

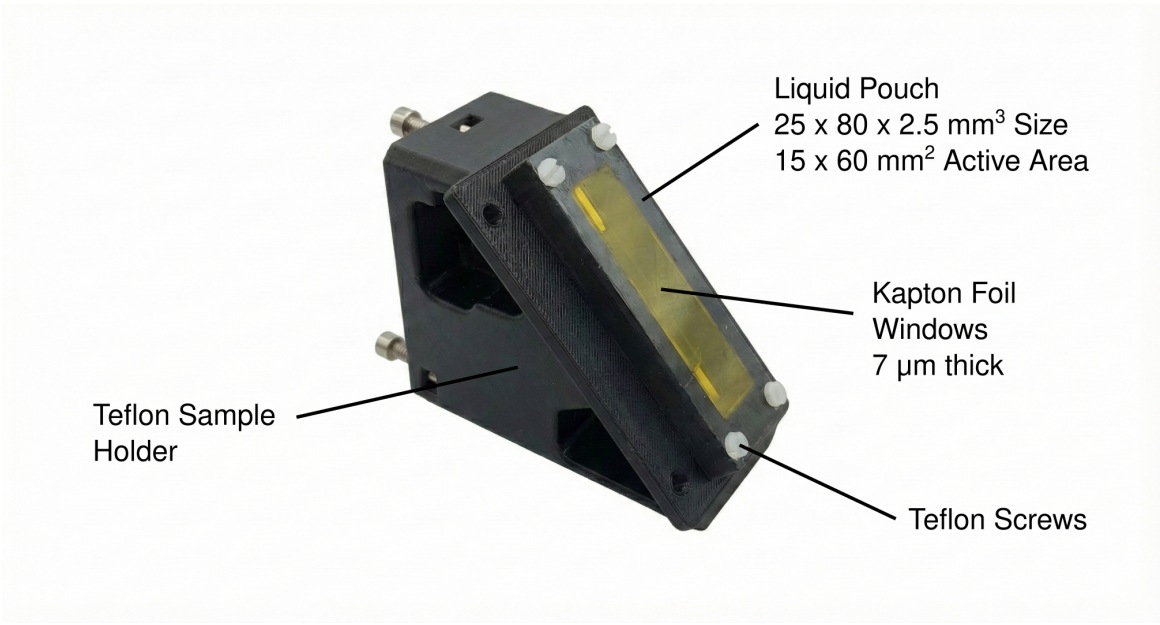


Fig. 4: Teflon liquid-sample holder used for solution measurements. The liquid is contained in a flexible pouch fixed with Teflon screws to prevent metal contamination, and sealed with a Kapton window to ensure X-ray transmission.

Conclusion and Outlook

In this work, we presented the recent developments of the VOXES spectrometer at the INFN National Laboratories of Frascati. The setup now features full motorization of all key components, including slits, crystal and detector, ensuring stable, reproducible, and remotely controlled operation for laboratory X-ray spectroscopy. The implementation of a dedicated sample holder, with the possibility to perform measurements on liquids, expands the range of measurable materials to include solutions and non-solid samples. In addition, the integration of a complementary ED-XRF line based on a silicon pin-diode enables fast alignment, flux monitoring, and simultaneous WD+ED acquisition, significantly improving workflow efficiency. The mechanical layout has also been upgraded to allow operation in a transmission mode geometry. Through the introduction of a Y-shaped sample holder and a reversible tube mount, the system is now compatible with transmission mode measurements for acquiring XAS spectra. These improvements increase the flexibility and usability of the VOXES spectrometer, reinforcing its role as a compact and versatile platform for laboratory-scale XRF measurements. Future work will focus on completing the XAS implementation and further optimizing the automation and data-acquisition framework.

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Article

Enhancing Performances of the VOXES Bragg Spectrometer for XES Investigations

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Abstract: Utilizing a dispersive crystal for X-ray Emission Spectroscopy (XES) significantly enhances the energy resolution when compared with spectroscopy performed with just silicon drift detectors. This high resolution is particularly valuable for studying metals, as it offers essential insights into their electronic structures and chemical environments. Conducting such experiments in the laboratory, as opposed to synchrotron light sources, presents challenges due to the reduced intensities of X-ray tubes and, consequently, low signal rates, with the effect of increasing the acquisition time. In this study, we demonstrate that XES spectra can be acquired within a few hours for a CuNiZn metallic sample alloy while still maintaining a good energy resolution and a large dynamic range. This is achieved with the VOXES spectrometer, developed at INFN National Laboratories of Frascati (LNF), along with a background reduction procedure that enhances the signal from emission lines under study. This study is a showcase for improving the efficiency of XES in tabletop setup experiments.

Keywords: XES; mosaic crystal; HAPG

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1. Introduction

Light–matter interaction at X-ray energies can manifest through various mechanisms [1], offering diverse insights into the material under examination. Specifically, the absorption and emission of X-rays can provide similar or complementary information about the elements present within the sample and their properties.

High-Resolution X-ray Emission Spectroscopy (XES) [2] has been established as a technique for extracting valuable information about atoms and their chemical surroundings. XES is particularly valuable for metals, especially transition elements, as it does not only indicate their presence in the material but also provides information regarding their valence spin state [3], coordination chemistry [4], and ligand properties [5].

For instance, the core-to-core emission from the 2p to the 1s state, the Kα line, based on its strong fluorescence yield, is a standard indicator for extracting information about a metal's presence. Further insight can be extracted from the Kα line [6], like the spin–orbit coupling and Auger effect. The emission from the 3p to the 1s, the Kβ, represents a richer source of information about the metal's state. Indeed, due to the exchange splitting interaction between the 3p and 3d electrons, it exhibits a stronger sensitivity towards both the spin states and the chemical coordination of the metal. This information can be deduced by analyzing the complex and intricate lineshape of the Kβ, which is a combination of several peaks, as a result of the correlation between electrons. In the simplest scenario,

the lineshape can be decomposed into two primary peaks: the main peak, the K$\beta_{1,3}$, and a shoulder, the Kβ', about 10 eV below the K$\beta_{1,3}$. The latter determines the spin sensitivity of the Kβ, as its prominence relative to the lineshape is sensitive to the number of unpaired electrons and, therefore, to high or low spin configurations of the metal [7]. At higher energies, close to K$\beta_{1,3}$, an additional peak is present: the K$\beta_{2,5}$, characterized by reduced intensity. It is determined by valence to core transitions and hence is strongly influenced by the chemical coordination of the metal.

XES's possibility to probe the local electronic structure and magnetic and chemical properties of the chemical element finds several applications in various fields, particularly in the characterization of metals, for instance, in coordination chemistry [8], where this method plays an important role in explaining the electronic states of metal complexes and understanding the coordination environment around metal ions. Additionally, in materials science, it aids in the comprehensive characterization of metallic materials, while in catalysis [9], it helps in understanding the intricate electronic structures of metal catalysts, providing crucial insights into their reactivity and catalytic mechanisms.

In XES experiments, it is required to excite the sample and induce the transition of electrons down to deeply bound core states. The initial excitation can be achieved, for instance, using X-rays, electrons, or protons. The typical X-ray sources are the synchrotron light ones, which, thanks to their large intensity over a broad energy range, enable the acquisition of XES spectra in a reduced amount of time. In recent decades, there have been a growing number of attempts to enhance the efficiency of such experiments when conducted in the laboratory. In this context, the source is an X-ray tube [10] that possesses a lower effective intensity, as most of the power is dissipated as heat, which results in an increased acquisition time.

Detecting the emitted X-rays is also crucial in order to extract the properties mentioned before. In this context, an extremely good resolution is needed to resolve the emission lines. For this purpose, an energy resolution at a few eV level is required at keV energies. Silicon Drift Detectors (SDD) [11], which possess an intrinsic resolution (FWHM) of about 120 eV at 6 keV, are not able to finely resolve emission lines but only distinguish the Kα and Kβ for a given metal. Transition Edge Sensors [12], instead, can be used to finely resolve the emission line based on a few eV resolutions at 6 keV [13]. The drawback of such detectors is their high cost due to the required operating temperature of about 50 mK.

Alternatively, exploiting Bragg's law to monochromate the emitted X-ray with a crystal is the standard approach to obtain high-resolution XES spectra. Various crystal geometries and combinations of crystals have been employed, such as Von Hamos [14] and Johann/Johansson configurations [15]. Additionally, utilizing mosaic crystals [16] offers the possibility to extend the energy's dynamic range for the XES spectra. Due to the presence of microcrystallites with different crystal orientations, a mosaic crystal can deflect a range of X-rays around the selected energy; this effect is quantified by their mosaicity value. The critical aspect of employing mosaic crystals is the nontrivial calibration needed to relate the energy of the emitted X-ray to its position in space on a strip detector or on the hit pixel for a CCD [17].

Efforts to enhance the efficiency of XES measurements in the laboratory have been pursued with the VOXES spectrometer, developed and optimized at the National Laboratory of Frascati (LNF) of INFN [18–20]. The setup is using an X-ray tube and HAPG crystal in the Von Hamos configuration, which makes it possible to acquire XES spectra for both solid and liquid samples in a few hours, thanks to the possibility of effectively shaping the emitted X-rays with two slits. This has the effect of increasing the effective source size up to the order of mm and enhancing the overall signal rates. It's important to note that there is always an intrinsic trade-off between the collected intensity and energy resolution, which varies depending on the required application. With such improvements, VOXES still maintains an energy resolution below 10 eV in the range of 4–12 keV.

In this work, we showcase the potential of utilizing the VOXES setup to efficiently acquire XES spectra in a reduced amount of time. As a demonstration, we discuss the

spectrum of a CuNiZn alloy, captured in one hour of acquisition time. This represents a significant reduction in the typical duration required for a tabletop setup, especially considering that peak rates in these experiments usually fall within the Hz range. The spectrum has a large dynamical range of 600 eV and a good energy resolution (FWHM) of less than 10 eV at 8 keV, allowing the simultaneous examination of multiple emission peaks from the three distinct elements of the alloy. Such properties are possible thanks to a refined background reduction procedure applied to the spectrum.

We show that we can effectively clean the spectrum with this procedure, enhancing the signal-to-background ratio of the emission lines under investigation. We then discuss the energy calibration and further explore the possibility of calibrating the spectrum without the inclusion of the nickel line.

The paper is structured as follows. Section 2 provides details about the experimental apparatus and the spectrum acquisition. In Section 3, the spectrum of the CuNiZn alloy is first presented. Subsequently, the procedure for background reduction is outlined, together with the determination of the calibration function and a comparison with the reference energy values of the emission lines. Section 4 concludes the paper and outlines possible future directions for the employed methods.

2. Setup and Methods

For the measurements reported in this paper, we utilized the VOXES spectrometer, as already discussed in our earlier works [18,19]; additional details can be found in those references. As depicted in Figure 1, VOXES operates in air and it consists of four main components. A source box accommodates the sample and houses an OXFORD XTF-5011 X-ray tube, whose anode is made of tungsten. A slit system with two STANDA slits, denoted as S_1 and S_2, is used to shape the X-rays from the source box, required to enlarge the effective source size, which impacts the resulting energy resolution and recorded rate of the spectrum. For this study, the selected values for S_0' and angular acceptance $\Delta\theta'$ were 1.0 mm and 0.4°, respectively, as defined in the referenced literature. A HAPG mosaic crystal made of pyrolytic graphite with a declared mosaicity of 0.1 ± 0.01, a thickness measuring 100 µm, and a curvature radius ρ_c = 206.7 mm was used. Finally, a MYTHEN2 strip detector, manufactured by DECTRIS, with an active area of 32×8 mm^2, equally divided into 640 strips, each with a depth of 450 µm and 50 µm pitch, was used for X-ray detection.

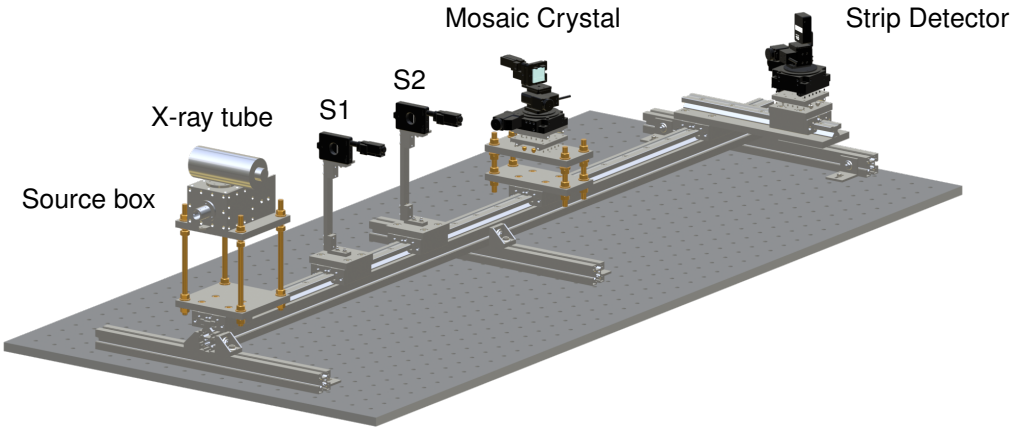


Figure 1. The VOXES setup. The experimental apparatus consists of an X-ray tube mounted on a source box where the sample is placed, two motorized slits, S1 and S2, a HAPG mosaic crystal, and a strip detector.

Movements for aligning the crystal and the detector to the specific emission line angle θ_B are facilitated by FESTO electrical motors. For more precise adjustments, a 5-axis system from STANDA controls the positioning of both the crystal and detector. The CuNiZn

sample used in this experiment is produced by Goodfellow and consists of 64% Cu, 24% Ni, and 12% Zn. It has dimensions of 25 × 25 mm and a thickness of 100 µm.

During the spectrum acquisition, the X-ray tube operated with an applied current of 500 µA and a voltage of 20 kV. The acquisition lasted one hour and was divided into one-second frames, during which counts over the detector strips were collected. The frame duration was optimized to ensure a sufficient number of events to allow the background reduction procedure described in the next section.

3. Results and Discussion

In this section, we present the results obtained for the CuNiZn alloy. The procedure for the background reduction is outlined, together with the description of the methodology used for calibrating the spectrum.

3.1. Background Reduction

The CuNiZn XES spectrum exhibits three principal emission lines (see Figure 2), namely the copper's Kα, the nickel's Kβ, and zinc's Kα. The spectra before and after background reduction are displayed in Figure 2, where the data from 640 strips are rebinned four times and the fit is performed on the background-subtracted spectrum. Due to the resolution at FWHM below 10 eV at 8 keV of our configuration, the Kα lines are well-approximated with two Gaussians for Kα_1 and Kα_2. For the Kβ line of nickel, we utilized a sum of two gaussians for the K$\beta_{1,3}$ and Kβ' peaks. The latter is justified by the emerging asymmetric profile of the Kβ at lower energies.

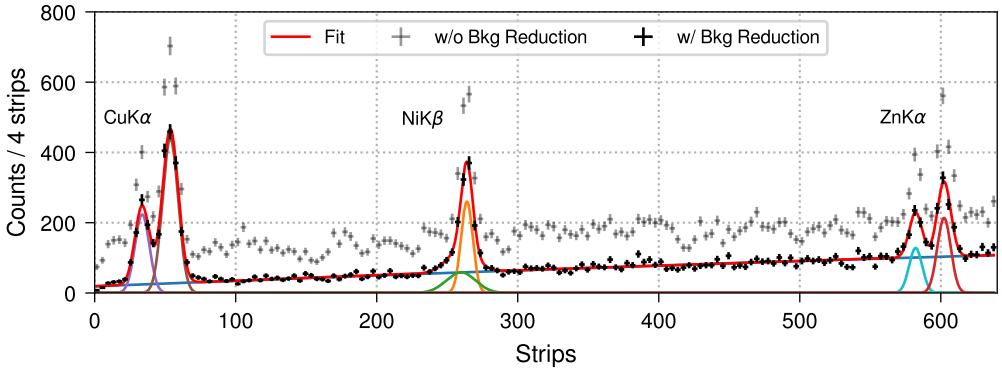


Figure 2. Spectrum of CuNiZn. The XES spectrum of the CuNiZn sample, with background reduction depicted in black and without it shown in gray; various colors are used to illustrate different Gaussian contributions for each emission line.

The idea behind the background reduction stems from the fact that three types of events are possible during the acquisition: from emission lines of interest, specifically from the lines of the elements of the alloy, from random events caused by thermal fluctuations or from cosmic rays, which are consistently present, and from artifacts in the electronics of the detector, such as crosstalk among different strips.

Here, we outline a procedure specifically designed to clean the spectra, aiming to preserve the expected physical properties of the emission peaks. During the hour-long acquisition, we analyzed the distribution of Total Counts per Frame (TCF) across the strips for each frame, identifying a main peak and a tail, as depicted in Figure 3. By selecting only those frames where the number of counts in the spectrum exceeds a certain threshold, we effectively reduced the background. This is demonstrated in Figure 2, where the raw spectrum is shown in gray, and the spectrum with a TCF threshold set to 10 appears in black. This approach results in a minimal reduction in the overall intensity of the peaks of interest, while effectively enhancing the signal from the emission peaks relative to the background.

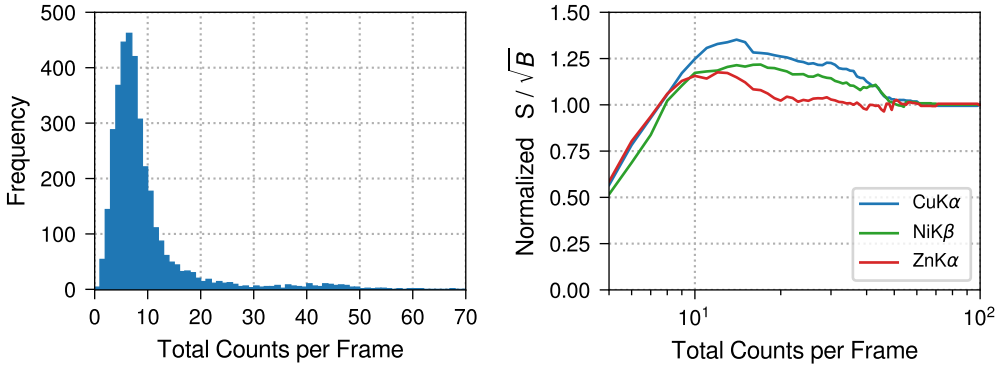


Figure 3. TCF distribution and background reduction effect. Distribution of the Total Counts per Frame (TCF) over the detector's strip during the acquisition (**left**). On the (**right**): the effect of selecting frames with a threshold of TCF distribution for the signal to background ratio, normalized to the value before the reduction procedure.

To further illustrate this procedure, we performed a fit for each TCF value, as shown in Figure 3. Here, the ratio of the number of events for the three lines over the background events is plotted as a function of TCF. For clarity, signal-to-background ratios corresponding to the same emission line are combined (i.e., $K\alpha_1$ and $K\alpha_2$ to $K\alpha$), and the ratios are normalized to the value without a threshold. Notably, the application of a TCF threshold leads to an immediate increase in these ratios, indicating that most of the background events are effectively removed. For low values, there's an observable enhancement in the ratios before they decline at smaller TCF values, where events are removed more uniformly across the spectrum. To identify the optimal threshold value, as detailed in the following section, we chose the value that, after the energy calibration, aligns the peaks more closely to their reference values in the literature. For an unknown sample with unidentified elements, one approach could involve fitting the distribution of total counts and selecting frames that fall within 1 or 2 standard deviations (sigma) from the distribution's mean.

3.2. Energy Calibration

The calibration of XES spectra was largely discussed in our previous works [19]. In this current study, we refined the calibration function, and we conducted further calibrations to determine the optimal threshold value for the background reduction procedure. This involved finding the threshold value that reduces the difference between the energy positions of the calibrated peaks and their reference values.

The geometry required for defining a calibration function is depicted in Figure 4, illustrating all the key elements of the VOXES setup: an extended source, a mosaic crystal with curvature radius ρ_c, and a position strip detector. Photons matching the exact energy of the line of interest possess the correct Bragg angle θ_B, and they are deflected to hit the detector at the nominal position $S(\theta_B)$. In this case, we focused on the $K\beta$ of nickel, which results in a $\theta_B = 12.75°$. Due to the mosaic nature of the crystal, photons without the proper orientation are also impinging on the detector. The position of a general photon on the detector is denoted by $S(\theta)$.

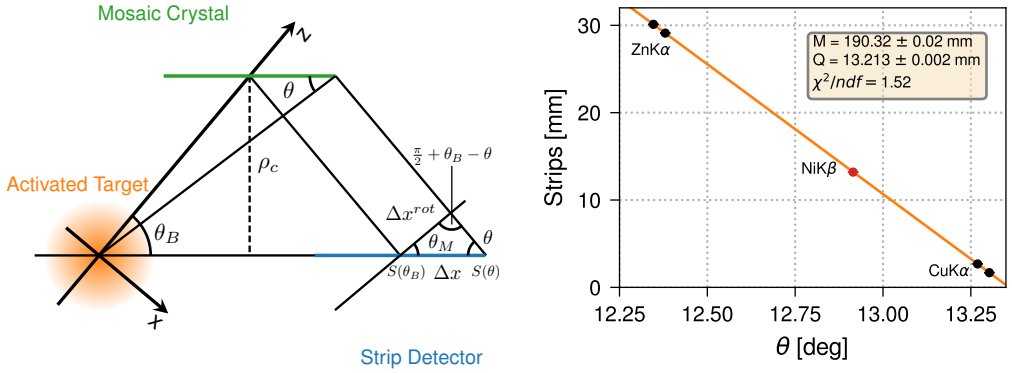


Figure 4. Geometry of the configuration and the calibration function. Schematization of the geometry (not in scale) employed for the VOXES setup (**left**) and the fit for determining the calibration function (**right**) without the inclusion of the nickel Kβ. The peak's position extracted from the fit for the Kα lines is in black and the position of the nickel's Kβ, not included in the fit, is in red.

Exploiting the fact that these positions on the detector, as a function of the curvature radius of the crystal and the angle of deflection, are $S(\theta_B) = 2\rho_c \cot\theta_B$ and $S(\theta) = 2\rho_c \cot\theta$, the difference between the distances on a plane parallel to the crystal is Δx, and it is equal to

$$\Delta x(\theta) = 2\rho_c(\cot(\theta) - \cot(\theta_B)) \tag{1}$$

where we adopted a definition for the sign of Δx to establish the correct inverse proportionality between the distance Δx and the angle θ. On the rotated plane where the detector is positioned, this distance is transformed (utilizing the law of sines) to

$$\Delta x^{rot}(\theta) = \frac{[2\rho_c(\cot(\theta) - \cot(\theta_B))]\sin(\theta)}{\cos(\theta - \theta_B)} \tag{2}$$

The latter equation relates (and unequivocally identifies) the angle of the photon to the position on the detector. The conversion from angle to energy is then performed with the usual Bragg's law, $\lambda = 2d\sin\theta$, where, for the mosaic crystal, $d = 3.356$ Å. To use such a formula for our case, two modifications are needed. First, there should be an offset term because we positioned our detector to have all three lines in the spectrum. This has the effect that the distance $\Delta x^{rot}(\theta)$ is not zero for $\theta = \theta_B$. Another aspect is related to the alignment of the experiment. We rotated the crystal around the nominal angle θ_B^{Ni} and the detector perpendicular to that, but due to the non-linear behavior of the mosaic crystal, even small misalignments produce large displacements in the position of X-rays on the detector. To account for this effect, we introduced a scaling parameter M in the calibration function, which takes into account this aspect. The employed calibration function is then

$$\Delta x^{rot}(\theta; M, Q) = \frac{[2M(\cot(\theta) - \cot(\theta_B))]\sin(\theta)}{\cos(\theta - \theta_B)} + Q \tag{3}$$

The scaling parameter M and offset Q have units of measurement in mm, and we determine their values with a fit between the reference energies (expressed as angles) of the peaks and their position on the detector. This process involves converting values from strips to millimeters, with the understanding that each strip measures 50 µm. The reference energies for the emission lines were taken from xraylib [21], namely $E^{Cu}_{K\alpha 2}$ = 8027.9 eV, $E^{Cu}_{K\alpha 1}$ = 8047.8 eV, $E^{Ni}_{K\beta 1,3}$ = 8264.7 eV, $E^{Zn}_{K\alpha 2}$ = 8615.8 eV, and $E^{Cu}_{K\alpha 1}$ = 8638.9 eV.

We performed two types of calibrations, with and without the inclusion of the nickel Kβ line in the fit. This was done because the K$\beta_{1,3}$ position may vary in general, as was largely discussed in the first section, and it makes it not really suitable for the determination of the calibration function. In the end, we compared the energies of the peaks with the reference values using the two approaches to estimate the systematic errors for not including the nickel Kβ line.

As the two functions are nearly indistinguishable, we report only the fit of the calibration function for the case without the inclusion of the nickel Kβ, as shown in Figure 4. From the fit, just using the Kα line's peak's position (black dots), it is possible to recover the Kβ position (red dot). For the offset term Q, we recovered the position in space of the nickel Kβ, as expected. For the scaling parameter M, we found the value of 195.32 $\pm$ 0.02 mm. This value is, remarkably, within 5% of the crystal's nominal radius of curvature ρ_c, a measure of the fact that our configuration is almost perfectly aligned.

To validate the accuracy of the approach, we computed residuals between the reference energy peaks and the center position by fitting the calibrated spectra. In Figure 5, the residuals before and after the background reduction are reported. The threshold value employed to filter the frame was 10 counts, as it gives the smallest residuals, and, with this value, there is still a gain in the signal-to-background ratio. The effect of the background reduction is to center the residuals around zero, but, most importantly, to reduce, on average by 30%, the uncertainties over the residuals and, therefore, over the position of the peak. The difference is also qualitative, as the background reduction is also able to reveal the asymmetric structure of the Kβ of nickel. Additionally, we also calculated the residuals for the case where the Kβ of nickel is not included, and the residuals are still within 0.1 eV with respect to the case where all the lines were included.

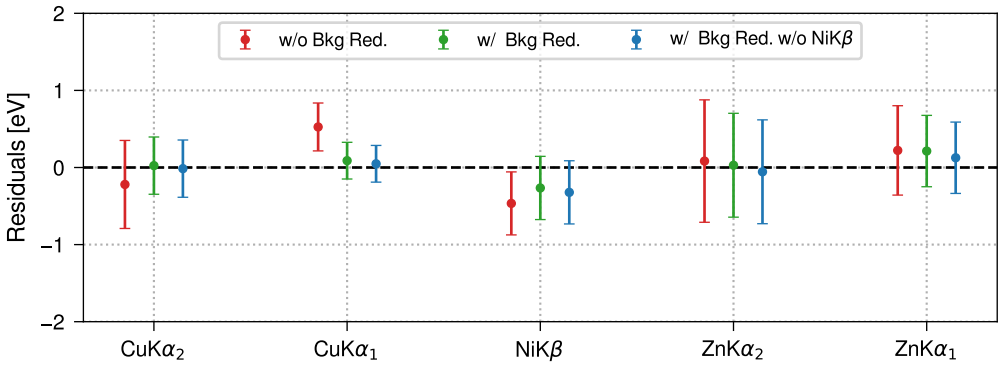


Figure 5. Effect of background reduction on residuals. Residuals obtained from the fit of the calibrated spectra without (red) and with the background reduction (green). The case with background reduction and without the inclusion of the nickel's Kβ in the calibration fit (blue) is also reported.

4. Conclusions and Outlook

In this paper, we reported a combined procedure that allows for the obtaining of high-resolution XES spectra with a reduced acquisition time. This was done by developing a background reduction procedure that permits cleaning the spectra and enhancing the emission peaks under examination. We used Bragg spectroscopy with the VOXES setup at INFN LNF, and we showcase the emission lines from a CuNiZn alloy. In the region of interest of our spectrum, there are the Kα lines of copper and zinc and Kβ of nickel. The background reduction was made possible by analyzing the distribution of the total number of counts for each frame of the acquisition. We demonstrated that applying a threshold to the total counts per frame qualitatively cleans the spectrum. This is clear from the reduced background noise and from the fact that it reveals the typical asymmetric structure of the Kβ line of nickel. We also measured the effect of the background quantitatively by calculating the signal-to-background ratio of the emission peaks as a function of the total number of counts for every frame. The ratios increased when frames with counts below a certain threshold were selected, and we determined an optimal threshold of 10 total counts per frame, as it minimizes the residuals between the calibrated peaks and their reference values. This procedure further enhances the precision of the calibrated spectrum by reducing, on average by 30%, the uncertainties associated with the peak centers of the emission lines, compared with the spectrum obtained without background reduction.

Further improvements are possible. For instance, adjusting the two slits can improve the energy resolution, despite reducing the emission rate and therefore increasing the acquisition time. This approach could even lead to fully resolving the lineshape by achieving precision below the intrinsic linewidth of the peaks. Further aspects to explore include refining the parameters for the calibration function to maintain a parameter-free calibration function. These refinements would significantly improve the consistency and reproducibility of our results and show the utility of XES in tabletop experiments for exploring the emission lines of metals.

Author Contributions: Conceptualization, A.S.; methodology, S.M., D.S., A.C., K.P., G.M. and A.S.; software, S.M., K.P., A.C., D.S. and F.N.; formal analysis, S.M., F.N., G.M. and D.S.; investigation, S.M. and A.S.; writing—original draft preparation, S.M.; writing—review and editing, S.M., F.N., C.C. and A.S.; supervision, A.S., C.C. and F.N.; project administration, A.S. and C.C.; funding acquisition, A.S. and C.C. All authors have read and agreed to the published version of the manuscript.

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Data Availability Statement: The data presented in this study are available on request from the corresponding author.

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Conflicts of Interest: The authors declare no conflicts of interest.

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Machine-learning-enhanced event selection for BEGe detectors in the VIP experiment

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Abstract: The VIP collaboration operates a Broad-Energy Germanium (BEGe) detector at the Gran Sasso National Laboratory to measure radiation in the few-keV to 100 keV range, aiming to search for spontaneous collapse-induced radiation and atomic transitions violating the Pauli Exclusion Principle. Here, we present a machine-learning-based upgrade for the BEGe detector of an event-selection strategy aimed at improving the efficiency in detecting low-energy events down to 10 keV. The method employs a denoising autoencoder to suppress electronic and microphonic noises and reconstruct pulse shapes, followed by a convolutional neural network that classifies waveforms as normal single-site events or anomalous events. The workflow was validated on a dataset comprising more than 20,000 waveforms recorded in 2021. The classifier achieves a receiver operating characteristic (ROC) curve with an area under the curve (AUC) of 0.99 and an accuracy of 95%. Applying this procedure lowers the minimum detectable energy of the final spectrum to approximately 10 keV. It also yields a measurable enhancement in spectral quality, including an improvement of about 14% in the signal-to-background ratio and improvement of the energy resolution for the characteristic Pb and Bi gamma lines. These developments enhance the sensitivity of the BEGe detector to rare low-energy signals and provide a scalable framework for future precision tests of quantum foundations in low-background environments.

Keywords: Data processing methods; Solid state detectors

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1 Introduction

Experimental tests of quantum mechanics foundations require extreme sensitivity, as they target phenomena that, if observed, would occur at extremely small rates [1]. Such rare-event searches provide a unique window into possible deviations from the standard quantum framework, complementing more conventional precision tests and opening opportunities to probe new physics at the interface between microscopic and macroscopic scales [2]. An example of such tests is the search for violations of the Pauli Exclusion Principle [3]. PEP is one of the most fundamental principles of quantum theory, which underlies the structure of atoms, the stability of matter [4], and even astrophysical objects like neutron stars [5]. Its validity is derived from the Spin-Statistics Theorem [6]. This theorem is intimately connected to locality and Lorentz covariance [7]; therefore, any violation of PEP would have profound implications, pointing to physics beyond the established local quantum field theoretical framework [8–10]. From an experimental point of view, such violations could manifest as anomalous atomic transitions, into already double-occupied states, shifted from the standard spectral lines [11]. Establishing increasingly stringent limits on such effects is therefore a central goal of dedicated rare-event searches. Another direction involves models of spontaneous wave-function collapse, which resolve the measurement problem by introducing additional physical mechanisms that suppress quantum superpositions at macroscopic scales. Prominent examples include the Continuous Spontaneous Localization (CSL) [12, 13] and the Diósi-Penrose (DP) [14, 15]. Both predict the spontaneous emission of radiation, in particular X-rays, which can serve as experimental signatures of the individual collapse model, especially in the low-energy regime (10-100 keV) [16].

Detecting such rare and subtle signatures requires not only detectors with excellent energy resolution but also an extremely low-background environment. Underground laboratories play a crucial role in this context. The Gran Sasso National Laboratory (LNGS) of INFN provides a uniquely low-background environment for the study of rare events. Located beneath approximately 1400 m of rock, a water equivalent (w.e.) to about 3800 m.w.e., the Gran Sasso overburden suppresses the cosmic muons by a factor of 10^6, i.e. a flux of 3.41×10^{-4} m^{-2}s^{-1} [17]. This substantial attenuation of the muon flux enables experiments at LNGS to achieve significantly reduced background levels. This makes LNGS one of the most favorable environments for rare-event searches, hosting experiments ranging from neutrino physics to dark matter and double-beta decay [18–21]. Taking advantage of this low-background environment, the VIP (Violation of the Pauli Exclusion Principle) collaboration has progressively tightened the upper limits on PEP-violating searches. The original VIP experiment set an upper limit on the PEP violating probability for electrons of $\beta^2/2 < 4.7 \times 10^{-29}$ [22], the

most stringent constraint at the time. Its successor, VIP-2, pushed this further, establishing limits of $\beta^2/2 \leq 6.8\times10^{-42}$ (Bayesian, 90% CL) and $\beta^2/2 \leq 7.1\times10^{-42}$ (Frequentist CL_s, 90% CL) [23]. The ongoing effort is extending the search across the periodic table, with high-Z targets such as zirconium, silver, palladium, tin [24] and germanium as well.

Reaching such limits requires detector technologies that can operate with high efficiency and stability in the keV range. As an example, Broad Energy Germanium (BEGe) detectors operating at LNGS, combine the features of coaxial and low-energy high-purity germanium designs, covering a wide dynamic range from a few keV up to several MeV [25]. They are particularly suited for rare-event searches because of their excellent resolution and efficiency at low energies [26]. Achieving energy thresholds of a few keV is technically demanding, as microphonic noise and electronic instabilities tend to obscure faint signals precisely in this region. Access to this low-energy window is essential: it is where the predicted spectra of different collapse models exhibit their strongest discriminating power [16], and it is also the key region of interest for PEP-violation searches, since the normal Kα line of germanium lies at 9.8 keV. In recent years, machine-learning (ML) techniques have emerged as powerful tools for improving pulse-shape analysis [27, 28] and low-energy event selection in germanium detectors, enabling discrimination between single-site and multi-site interactions [29] and identification of event classes relevant for rare-event searches such as neutrinoless double-beta decay [30]. A practical challenge that remains is the consistent recovery of low-energy signals for BEGe detectors, which carry much of the sensitivity for both PEP and collapse-related measurements [31].

In this work, we report recent advances in the operation of a BEGe detector at LNGS. We developed and validated a ML-based event selection strategy, employing a denoising autoencoder (DAE) and a convolutional neural network to enhance discrimination between normal and anomalous events. These methods were applied to data acquired in 2021, demonstrating improved effective energy threshold and a higher signal-to-background ratio. The paper is organized as follows. Section 2 introduces the ML framework for event selection, with emphasis on the denoising and classification stages. Section 3 presents the performance validation on the 2021 dataset and discusses the improvements in resolution, threshold, and background rejection. Section 4 concludes with a summary and an outlook for future developments within the VIP program and related rare-event searches.

2 Event selection strategy

Event selection was performed using pulse shape analysis (PSA) on the acquired waveforms of the BEGe setup. To automate this selection and integrate it in the data acquisition, we employed ML with a DAE [32] with two main objectives. First, to remove noise from the waveforms and improve the selection of low-energy events. Second, to extract features that help in distinguishing normal events from anomalous ones (e.g. multi-site, saturated, and similar events). The latter is important for the creation of a dataset for supervised training of a classifier connected to the output of the DAE, allowing for anomalous detection of events.

This ML-aided PSA procedure was validated using the data collected in 2021, consisting of more than 20,000 events. For each event, a waveform is recorded by the BEGe detector electronics through a 400 MHz digitizer spanning 1024 samples, with a total acquisition window of 2.55 µs. Further details on the BEGe apparatus are provided in our previous article [31]. We implemented the DAE within the Keras framework (v3.5.0). The DAE, which operates on both the waveform and its derivative, is based on a one-dimensional (1D) convolutional neural network (CNN) with an

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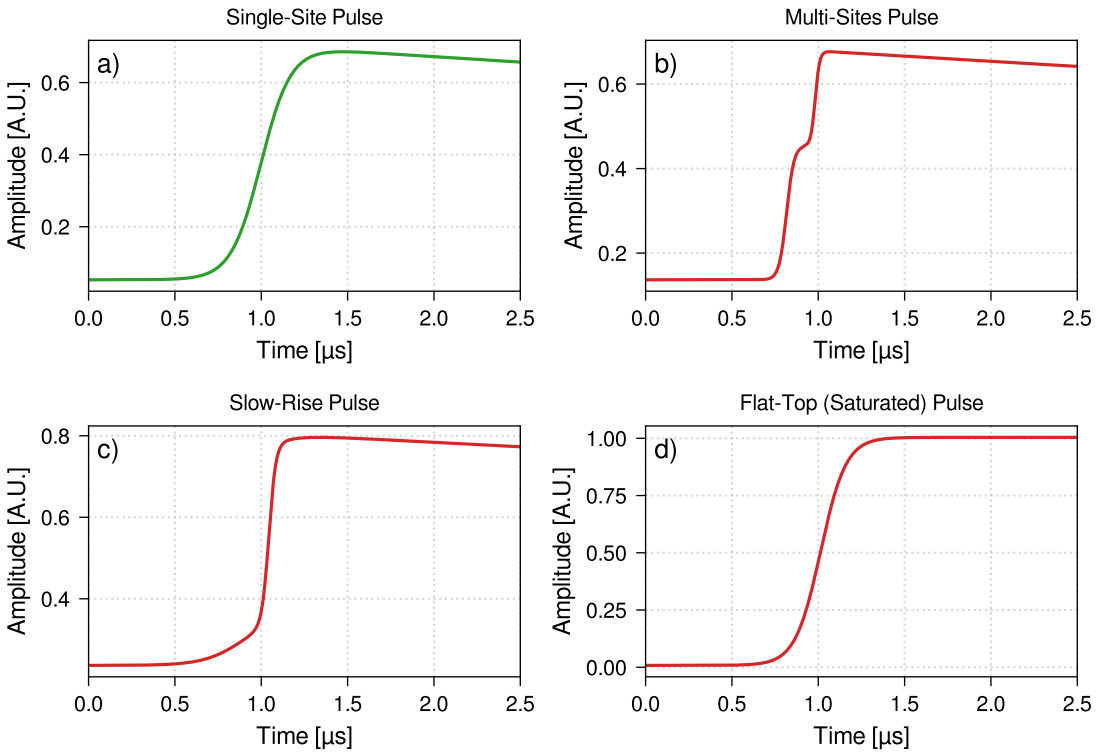


Figure 1. Examples of the four pulse types in the synthetic dataset used to train the DAE: a normal single-site pulse (green) in panel a, and anomalous pulses (red): multi-site (panel b), slow-rise (panel c), and saturated (panel d).

encoding and decoding structure. The encoder comprises three convolutional layers with progressively decreasing filter sizes (i.e. 64, 32, 16), each followed by the rectified linear unit (ReLU) activation function, and max-pooling operations. The temporal dimension of the waveform in the input layer was reduced to 128 to decrease network complexity and shorten training time. The chosen input size and number of hidden layers represent a balance between model complexity, training efficiency, and the performance of both the DAE and the attached classifier. The decoder mirrors the encoder structure, using upsampling layers and additional convolutional layers with ReLU activation to restore the original waveform, followed by a final convolutional layer with linear activation to produce the reconstructed signal. The DAE is trained using the Adam optimizer [33] with a learning rate of 10^{-4}. No hyperparameter tuning was required, since both the DAE and the CNN classifier achieved excellent performance with their initial configurations.

To train the DAE, we created a synthetic dataset of pulses that reproduce the typical scenarios of events in our BEGe detector setup (see figure 1). A normal event (see figure 1(a)) corresponds to single-site energy depositions occurring within the bulk of the germanium detector, where the electric field is uniform and charge carriers are efficiently collected at the electrodes. These events are characterized by a prompt, steep rising edge and an exponential decay, and represent the expected signal. Multi-site pulses (see figure 1(b)) are generated to mimic the response to events in which energy is deposited at multiple, spatially separated locations within the detector volume. Such events can arise from, for example, Compton scattering or coincident background interactions. The resulting pulse shapes typically exhibit multiple rising steps or a more complex rise profile, reflecting the superposition of charge collection from each interaction site. Slow-rise pulses (see figure 1(c)) are

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included to model interactions occurring near the detector surfaces, where the electric field strength is reduced. In these regions, charge carriers drift more slowly to the electrodes, resulting in a notably slower rise-time compared to bulk events. The inclusion of slow-rise pulses in the training set allows the DAE to learn features associated with surface backgrounds, which can be subsequently suppressed through pulse shape discrimination, thereby improving the overall signal-to-noise ratio. Flat-top pulses (see figure 1(d)) are also incorporated in the dataset. These pulses are characterized by a rapid rise followed by a plateau, which can occur when the detector response saturates, due to high-energy depositions or limitations in the readout electronics. Including flat-top pulses in the training set is crucial to ensure that the DAE can recognize and appropriately process saturated signals, which may otherwise be misidentified or improperly reconstructed. These categories follow the standard classification established in BEGe PSA studies [25, 26].

Each simulated pulse was superimposed with noise to reproduce the conditions of the experimental setup. Two noise components were considered: (i) white Gaussian noise, representing intrinsic electronic fluctuations, and (ii) colored noise generated by integrating a white noise process to emulate the low-frequency microphonic and environmental instabilities observed in the detector. The relative amplitude of the two components was tuned to match the baseline noise spectrum previously characterized for the BEGe setup [31]. Two distinct noise levels were applied, yielding a total of 5000 simulated events evenly divided between high- and low-noise conditions. This approach ensured that the DAE was trained across realistic noise environments, thereby improving its robustness when applied to experimental data.

All events are preprocessed prior to training through baseline subtraction and normalization by their amplitude. For each pulse, the baseline is first estimated as the mean of the initial 100 samples and subtracted from the entire waveform to remove DC offsets. The resulting signal is then processed with a trapezoidal filter [34], and the amplitude is extracted as the maximum absolute value of the filtered output beyond the filter response region. The baseline-subtracted pulse is then normalized by this amplitude, resulting in a unit-amplitude, baseline-corrected signal. This standardized preprocessing pipeline is applied uniformly to both synthetic and experimental datasets to ensure robust and consistent inputs throughout model training and evaluation. It is worth noting that without this baseline subtraction and normalization, the DAE fails to accurately reconstruct pulse shapes.

The reconstructed pulses produced by the DAE serve as inputs for the binary CNN classifier for the detection of anomalous events. Each event has two input channels: the normalized pulse taken from the DAE output, which needs no further normalization, and the normalized first derivative of the pulse scaled to the range $[0, 1]$. Using both the pulse and its first derivative as separate channels, allows the CNN to learn the relation between the pulse shape and small changes, such as multi-site events.

To train the classifier, we create a dataset of waveforms with labels, by selecting features from the output of the DAE and placing constraints on their distributions. The following features were used:

- Rise time: the time needed for the waveform to rise from 10% to 90% of its amplitude.
- Number of peaks in the first derivative of the waveform.
- FWHM time: the duration of the full width at half maximum of the derivative for events with only one peak in the derivative.
- The L1-norm of the waveform with respect to the average pulse.

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A visual inspection of the rise time and FWHM time is shown in figure 2.

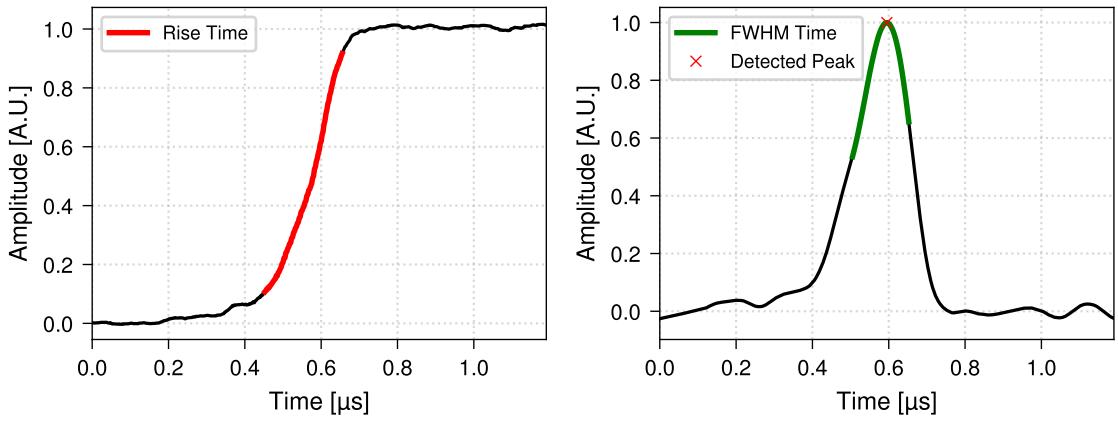


Figure 2. The rise time (left) is defined as the interval between 10% and 90% of the waveform amplitude (red). The FWHM time (right) is defined from the derivative of the waveform, with the FWHM interval highlighted in green.

To assess the validity of performing feature extraction on the DAE-processed waveforms, we evaluate the rise time and FWHM for all events before and after denoising, as illustrated in figure 3. The DAE does not change the features within $\pm\sigma$ of the main peak. This shows that the cleaning procedure does not distort the waveforms and supports the validation of the DAE transfer from synthetic data to real data. To further support this, we also show in figure 3 the mean squared error (MSE) of the DAE on real and synthetic data. The DAE gives similar values on both datasets, which indicates that it generalizes well to real data. The larger tail in the synthetic data is due to the different noise level used in the construction of the synthetic dataset. As an example, the effect of the DAE on a low energy event, where the reduction of the background is clear, is shown in figure 4.

Normal events for training are defined as those that satisfy all of the following conditions:

1. Feature constraints: the rise time, full width at half maximum (FWHM) time, and L1 norm each fall within one standard deviation (1σ) of their respective distributions (see figure 3).

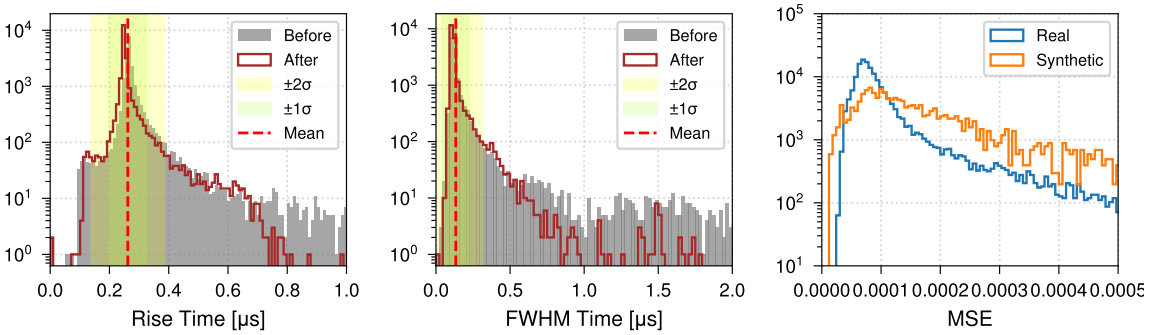


Figure 3. Rise time (left) and FWHM time (center) distributions for the events before and after applying the DAE, with mean and variance intervals. On the right is the mean squared error (MSE) of the DAE on the real and synthetic waveform data.

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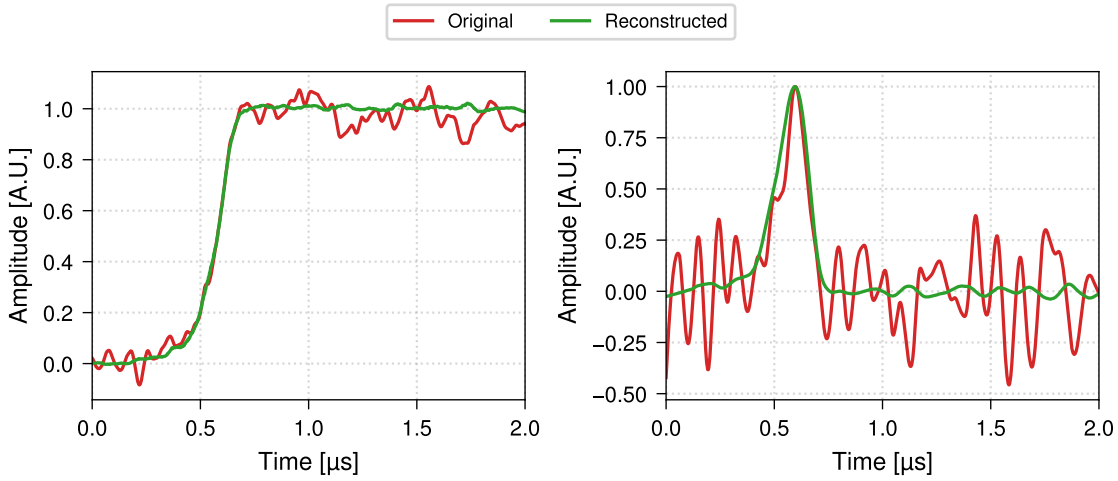


Figure 4. Effect of the DAE on a real waveform (left) and on its derivative (right) for a low-energy pulse which corresponds to an energy of ~ 15 keV.

2. Peak count: the event contains exactly one peak in the normalized first derivative, as determined with the `find_peaks` function from SciPy [35].

3. Labeling: all waveforms are manually inspected and labeled with a custom GUI program and assigned a normal or anomaly label.

The resulting dataset is already well defined by the first two criteria, and manual labeling changes the selection by only 3%. To train the CNN classifier, the dataset was split into training, validation, and testing with proportions 70%, 15% and 15%, respectively. The CNN model is composed of sequential 1D convolutional layers with increasing filter sizes (128,64,32) and decreasing kernel widths, each followed by batch normalization and max pooling to extract and condense features from the input. A fully connected dense layer interprets the extracted features, with a dropout layer included to mitigate overfitting. The final output layer uses a sigmoid activation function to produce a probability score for binary classification. The model is trained using the Adam optimizer with a learning rate of 10^{-5} and binary cross-entropy loss, and evaluated using accuracy, precision, and recall metrics. Early stopping of 10 epochs is employed to prevent overfitting and ensure optimal model weights.

3 Results and discussion

The DAE-CNN workflow was applied to the 2021 BEGe dataset to evaluate its performance and improvements on the final reconstructed spectrum and assess its suitability for integration into the online data-acquisition (DAQ) of the BEGe for VIP. After training, the CNN classifier assigns to each event the probability of being a normal (single-site) waveform. The classifier performance is quantified using the receiver operating characteristic (ROC) curve and the F1-score, as shown in figure 5, resulting in an area under the curve (AUC) of 0.99. The optimal operating point is defined by the threshold value (0.66) that maximizes the F1-score, corresponding to an accuracy of 95 %. At this working point, approximately 13 % of events are rejected as anomalous. A feature-based classifier trained on rise time and FWHM also provides good separation power, but computing these features online would introduce

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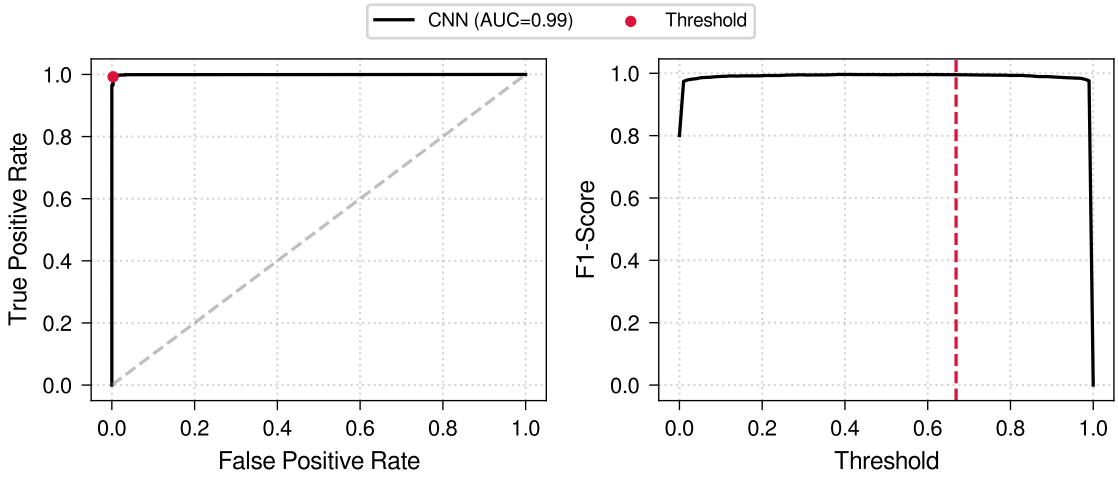


Figure 5. Performance of the CNN classifier applied to DAE-processed waveforms. Left: ROC curve with an AUC of 0.99, with the chosen operating threshold marked in red. Right: accuracy and F1-score as a function of the classification threshold, with the selected operating point indicated by the dashed red line.

significant computational overhead. In contrast, the CNN operates directly on the acquired waveform, maintaining high accuracy while remaining compatible with real-time inference, and allowing the DAQ system to store only classified results rather than full waveforms, thereby substantially reducing the data volume. Events with a classification probability above the optimal threshold are selected as normal (single-site) events. Spectra constructed by selecting events with the classifier can be divided as in figure 6. To extract the calibration, a fit with a binned maximum-likelihood approach was done. The calibration uses the four prominent gamma lines from the ^{238}U decay chain: ^{210}Pb at 46 keV, ^{214}Pb at 295 and 352 keV, and ^{214}Bi at 609 keV. The ^{212}Pb line at 238 keV is not included in the fit but used as an independent residual test. Each spectrum is fitted with the following model:

$$f(x) = e^{a_1+b_1x}\,\frac{1}{2}\left[1+\mathrm{erf}\left(\frac{x-\mu_1}{\sigma_1}\right)\right] + e^{a_2+b_2x}\,\frac{1}{2}\left[1+\mathrm{erf}\left(\frac{x-\mu_2}{\sigma_2}\right)\right] + \sum_i A_i\,\exp\left(-\frac{(x-\mu_{g,i})^2}{2\sigma_{g,i}^2}\right). \quad (3.1)$$

The background is modeled using two error-function components, chosen to be an empirical parameterization that provides a stable and reliable fit for the background. A physically motivated background model will be developed in future studies. The gamma lines are modeled with Gaussian functions. The fitted spectra and pull distributions are shown in figure 6. Using the DAE-CNN event selection, the effective energy threshold is lowered to approximately 10 keV, and the energy resolution at the 46 keV line is measured to be in the range of 1.0–1.5 keV (FWHM). The DAE significantly suppresses noise-induced distortions in the sub-15 keV region, enabling reliable amplitude extraction at low energy. Without the DAE preprocessing, the region below 15 keV is dominated by electronic noise and microphonic disturbances, limiting the depth of the analysis and reducing the sensitivity to low-energy processes. The DAE reconstructs physically plausible pulse shapes while suppressing noise components, significantly improving amplitude estimation and stabilizing the downstream classification. The ML-based selection enhances the signal-to-background ratio by approximately 14 %, with the largest improvement in the 10–50 keV region where anomalous waveforms contribute substantially to the background. Reaching an effective threshold around 10 keV provides full sensitivity

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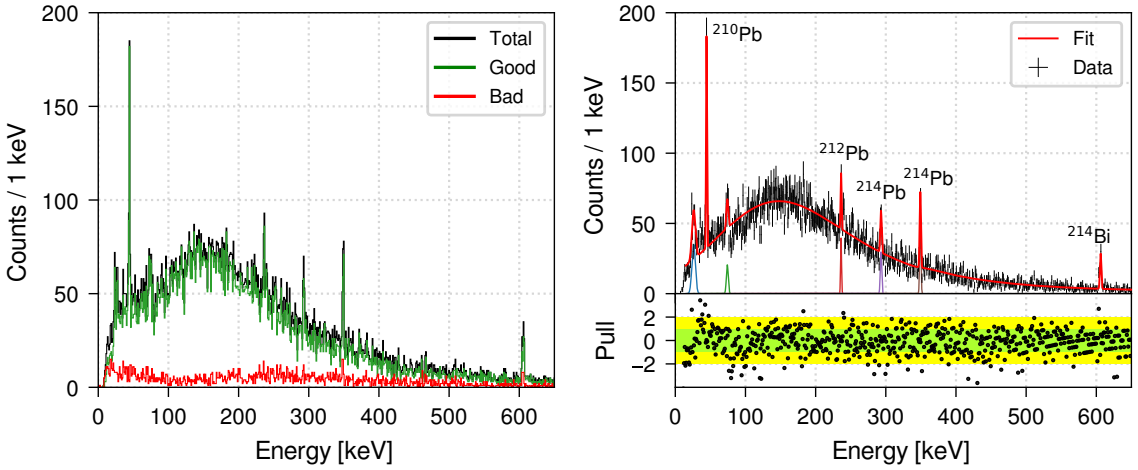


Figure 6. Left: measured spectra after ML-based event selection, showing normal (green), anomalous (red), and total (black) events. Right: fitted spectrum (red) with individual Gaussian components (coloured) and pull distribution below.

for collapse model tests, and with larger future data acquisitions it may become feasible to probe the vicinity of the 9.8 keV Kα line relevant for PEP-violation searches in germanium. Moreover, the DAE preprocessing enables alternative analysis strategies beyond the CNN classifier, including traditional pulse-shape discrimination and feature-based ML methods based on rise time, FWHM, or waveform derivatives. A Python package, `vipbege` [36], is provided together with this work, offering tools to extract the waveform features discussed here and to generate simulated waveforms with or without noise. This makes the approach transferable to future BEGe datasets and adaptable to similar low-background detector configurations.

4 Conclusions

In this work, we developed and validated a ML-based event-selection strategy for a BEGe detector operated by the VIP collaboration at LNGS. A DAE, trained on synthetic waveforms representative of the main pulse classes, was used to suppress microphonic and electronic noises and to extract robust pulse-shape features. A CNN classifier attached to the DAE output enables fast event selection and anomaly detection. Applied to the 2021 dataset, the method achieves an AUC of 0.99 and 95% accuracy, lowers the effective energy threshold from 20 keV to about 10 keV, and improves the signal-to-background ratio by 14%. These results demonstrate that ML-based waveform cleaning can significantly enhance the low-energy sensitivity of BEGe detectors, providing a valuable tool for studies of collapse-model signatures and future tests of the Pauli Exclusion Principle in low-background environments.

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Data Availability Statement. This article has associated data in a data repository. The labeled dataset of 2021 BEGe waveforms used in this study is publicly available on Zenodo under the DOI: https://doi.org/10.5281/zenodo.17641589.

Code Availability Statement. This article has associated code in a code repository. All scripts required to reproduce the figures presented in this article are available at: https://github.com/simonemanti/figures_bege.

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Precision test of bound-state QED at intermediate *Z* with kaonic neon

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We report Dirac-Fock calculations of transition energies for kaonic neon. For the most intense line, the 7–6 transition, the calculated energy is 9450.28 eV, which includes a bound-state quantum electrodynamics (BSQED) contribution of 12.66 eV. This is in excellent agreement with the recent SIDDHARTHA-2 measurement at DAΦNE of $9450.23 \pm 0.37\,(\text{stat.}) \pm 1.50\,(\text{syst.})$ eV. With the quantum electrodynamics contribution far exceeding experimental uncertainty, these results establish kaonic atoms as sensitive platforms for precision tests of BSQED in intermediate-Z systems.

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I. INTRODUCTION

Quantum electrodynamics (QED) is the most accurate theory describing the interaction between charged particles and photons within the framework of quantum field theory. For instance, in the context of bound-state QED (BSQED) [1,2], which utilizes QED to describe interactions in bound atomic systems, theoretical predictions achieve parts-per-billion (ppb) accuracy for the 2S–1S transition in hydrogen [3]. However, for intermediate and high atomic numbers (Z), the theoretical calculations face limitations in reaching similar precision, as the perturbative expansion scales with (αZ), with α the fine-structure constant, leading to convergence issues for high-Z elements. To explore the intermediate-Z case, tests of BSQED are typically conducted using highly charged ions (HCIs) [4–6] confined in cyclotrons and ion traps, enabling the exploration of QED effects in strong electric fields [7]. Nevertheless, for many experimentally accessible transitions, nuclear structure effects, such as the finite nuclear size (FNS), are comparable in magnitude to the higher-order QED contributions and can limit the achievable accuracy, particularly at intermediate Z [8]. Recently, a paradigm shift has occurred

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in BSQED tests at intermediate Z, favoring the use of exotic atoms [9]. These systems, where a negatively charged exotic particle (e.g., μ^-, $\bar{p}$, K^-, π^-) replaces an electron, enable probing regions closer to the nucleus due to the larger mass of the exotic particle compared to the electron, where BSQED effects are highly enhanced [10]. Historically, studies of exotic atoms were constrained to solid targets [11], where electron refilling effects often left residual electrons bound to the atom, complicating the interpretation of spectroscopic data [12]. Recent advancements in detection and accelerator technologies have enabled investigations using low-pressure gas targets, significantly reducing electron refilling [13–17] and allowing for cleaner observations of atomic transitions [18–20]. To date, BSQED tests in exotic atoms have predominantly utilized muonic, pionic, and antiprotonic atoms [9]. In contrast, negatively charged kaons present unique advantages for such studies [21]. Being spin-0 particles, kaons exhibit a strongly reduced hyperfine splitting, unlike muons, simplifying the spectral analysis. Additionally, having an intermediate mass between that of pions, muons, and antiprotons, kaons could provide for BSQED studies a more complete picture across the entire mass range. Moreover, specific transitions within the kaonic atom cascade can be selected to minimize the influence of FNS effects and residual electrons. A notable challenge in employing kaons for precision measurements is the "kaon-mass problem." Several discrepancies exist among measurements of the kaon mass [22–25], with the primary disagreement occurring between the two most precise results [26,27]. This has led to a persistent uncertainty of approximately 30 parts per million (ppm), as reported by the Particle Data Group (PDG) [28]. Resolving this issue is crucial for enhancing the accuracy of BSQED tests involving kaonic atoms. The majority of these studies were conducted prior to 1990, when accurate methods, particularly for addressing electron screening effects, were not yet well established. Nowadays, state-of-the-art multiconfiguration Dirac-Fock (MCDF) methods [29] offer the possibility to take into account all the different effects, potentially uniquely linking and determining the kaon mass from the experiment. Recently, the SIDDHARTA-2 Collaboration at the DAΦNE collider of the National Laboratories of Frascati (INFN-LNF) in Italy, in preparation for the subsequent measurement on kaonic deuterium, successfully performed high-precision x-ray spectroscopy measurements of kaonic neon (KNe with K for K^-) [18] using a gaseous target. This system exhibits several transitions with high yields (>30%) in the x-ray range corresponding to high-n levels. Thanks to the performed optimization and the use of silicon drift detectors (SDDs) [30], sub-eV precision was achieved on the measurement of those lines. In this paper we perform a more refined data analysis and compare the recent SIDDHARTA-2 KNe measurements [18] with state-of-the-art MCDF calculations. We demonstrate excellent agreement between experiment and theory, underscoring the sensitivity of KNe transitions, and kaonic atoms more broadly, to BSQED effects. We also validate the result against the electron screening effect and kaon-mass dependence, and finally show that KNe offers a viable route to refining the charged kaon mass with high precision.

II. METHODS

KNe data were recorded during the 2023 SIDDHARTA-2 run at the DAΦNE [31–33] collider at INFN-LNF. The sample used in this article corresponds to an integrated luminosity of 150 pb^{-1}, representing a 20% increase in statistics compared to our previous result [18]. Further experimental details on the SIDDHARTA-2 setup and KNe measurement are provided in Refs. [34] and [18], respectively. To extract the transition energies from the x-ray spectrum, we performed a maximum likelihood fit in which the observed spectral peaks were modeled using Gaussian functions. Each Gaussian function was explicitly parameterized by the SDDs resolution, incorporating both the Fano factor and electronic noise, consistent with the methodology described in our previous study [30]. Unlike the analysis in Ref. [18], here we explicitly account for the presence of different neon isotopes (^{20}Ne and ^{22}Ne) and model the lines 7i-6h, 8k-7i, and 9l-8k using a Gaussian function for each isotope. The amplitudes of the two components were constrained based on the natural isotopic abundances of neon [35], namely, a ^{22}Ne / ^{20}Ne ratio of 9.78%. The inclusion of ^{21}Ne, which has a natural abundance of 0.27%, as well as the uncertainty in the isotopic abundance ratios, does not affect the fit results within the sensitivity of this study. Systematic uncertainties on the experimental transition energies arise from both the energy stability over time and the calibration procedure, as previously described [36]. To compute the x-ray transition energies, we utilized the Multiconfiguration Dirac-Fock General Matrix Elements (MCDFGME) code (version 2025.1) developed by Desclaux and Indelicato [4,29,37], with the 2018 CODATA fundamental constants [38]. For each transition the code solves the Klein-Gordon equation for both initial and final states of the kaonic atom, and QED contributions are evaluated with vacuum polarization at all-order [39,40]. In all calculations, we consider only transitions between high-n circular states ($\ell = n - 1$). FNS effects are neglected by adopting the point-nucleus approximation, as their impact on transition energies is limited to a few meV, as reported in the Supplemental Material [41]. Recoil corrections are evaluated as in Ref. [42] and included to the transition energies. Electron screening effects are evaluated by including a single electron in the 1s orbital of Ne, constructing the total wave function from the electron and kaon. Unless otherwise specified, we adopt the reference kaon mass from the PDG [28] of $M_{K^-} = 493.677$ MeV. The MCDFGME calculations were used to further constrain the fit of the neon isotopic peaks: the centroid of the K^{22}Ne peak was fixed to the isotopic shift predicted by the calculation, similar to μNe [19], and the resolution of the peak was scaled accordingly. We validated this approach by assessing the change of the isotopic shifts for our lines of interest (7i-6h, 8k-7i, 9l-8k) to variations of the kaon mass w.r.t. the PDG value and found that changing the kaon mass by ±100 keV (more than seven times the current PDG uncertainty) leads to a shift of less than 10 meV (see Fig. 1) in the isotopic shift values. This confirms the robustness of our spectral modeling to fit different isotopic contributions.Radiative and Auger transition rates were calculated using the hydrogenic wave functions and standard analytical formulas widely used

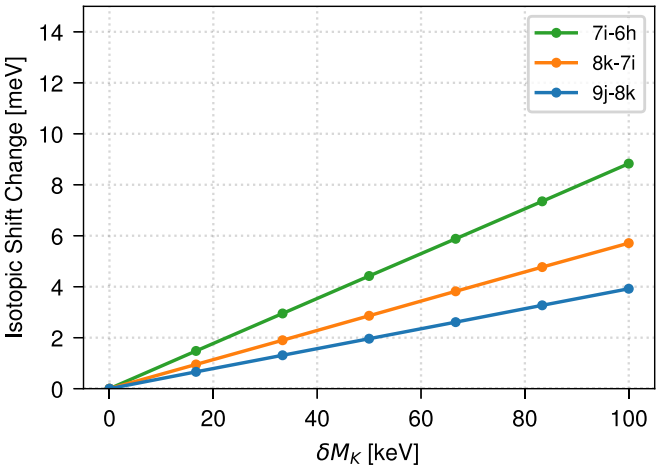


FIG. 1. Change of the isotopic energy shifts as a function of the kaon mass w.r.t. the PDG value for the 7i-6h, 8k-7i, and 9l-8k transitions in $K^{20}Ne$. The maximum variation over the ±100-keV range is 9 meV for 7i-6h, 6 meV for 8k-7i, and 4 meV for 9l-8k.

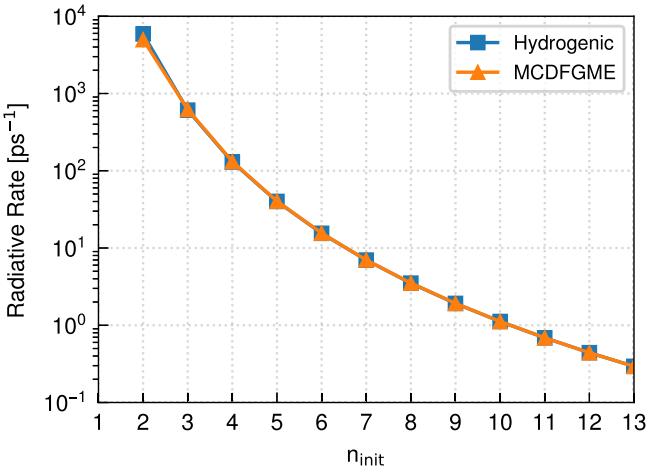


FIG. 2. Comparison of radiative transition rates between circular states in KNe calculated with the scaled hydrogenic formula [43] (blue) and by using MCDFGME wave functions (orange).

in exotic atom studies [43]:

$$\Gamma^{(R)}_{n,l\to n',l\pm1} = \frac{4\mu Z^4}{3}\alpha^3\left|R^{n,l}_{n',l\pm1}\right|^2(\Delta E_{if})^3$$

$$\Gamma^{(A)}_{n,l\to n',l\pm1} = \frac{16}{3}\left(\frac{Z_e}{Z}\right)^2\frac{\pi}{\mu^2}\frac{l}{2l+1}\left|R^{n,l}_{n',l\pm1}\right|^2 \times \frac{y^2}{1+y^2}\frac{\exp[y(4\tan^{-1}y-\pi)]}{\sinh(\pi y)}. \quad (1)$$

The Auger rate formula applies for an electron in the $1s$ orbital, and $Z_e = Z - 1$. ΔE_{if} is the transition energy for the hydrogenic case and $R^{n,\,l}_{n',\,l\pm1}$ is the dipole radial integral, μ denotes the reduced mass of the $K^{20}Ne$ system, and $y = \sqrt{k/2E}$ is the momentum of the ejected K-shell electron. While the development of a dedicated MCDFGME treatment for Auger transitions is planned for future work, we validate the hydrogenic approach by comparing dipole radiative rates from the hydrogenic approximation and MCDFGME calculations. In Fig. 2 the radiative rate from formula 1 is compared with the MCDFGME result for KNe, and for high-n transitions ($n > 5$) the relative error is found to be below 1%. Additionally, to further highlight the potential of KNe measurement, we extract the kaon mass directly from the measured x-ray transitions. This is accomplished by iteratively adjusting the kaon mass in the theoretical calculation until agreement with the experimental value is achieved, following the method of Gall *et al.* [27]. Here, we terminated the iterative procedure once the kaon mass change between successive steps fell below 0.5 keV.

III. RESULTS AND DISCUSSION

We now report the experimental KNe transition energies measured by SIDDHARTA-2 and compare them with the MCDFGME calculations. We then discuss the significance of these results for BSQED studies and their implications for a precise determination of the kaon mass.

A. KNe spectrum

The experimental KNe spectrum, shown in Fig. 3, features three dominant peaks corresponding to the 7–6, 8–7, and 9–8 transitions, observed at approximately 9.4, 6.1, and 4.2 keV, respectively. In addition, several lower-intensity peaks are present which include transitions with $\Delta n > 1$, such as 10–8, 11–9, 12–9, and 12–8. The spectrum also reveals x-ray lines from other kaonic atoms formed in the setup environment, including carbon, oxygen, nitrogen, and titanium. A distinct peak at 10.8 keV is attributed to the bismuth Lα transition, originating from bismuth in the SDD ceramic materials [30]. The identification and relative intensities of these contaminant lines are consistent with our previous observations [36], and their contributions have been carefully included in the analysis. The observed KNe transitions are consistent with expectations based on the competition between radiative and Auger decay channels during the cascade, as shown in Fig. 4. A qualitative analysis of the calculated rates predicts at which levels of the cascade radiative transitions begin to dominate over Auger processes. Specifically, for KNe, radiative decay becomes the dominant channel, starting from an initial principal quantum number of approximately $n_{\text{init}} = 9$ of the cascade (see Fig. 4). Transitions originating from levels below this threshold begin to appear in the spectrum, marking the progressive dominance with increasing yield of radiative emission over Auger deexcitation. A fit was performed to extract the transition energies of the observed peaks (see Fig. 3). The fitting procedure was refined relative to our previous study [18] to enable a direct comparison with theory, specifically by decoupling the contributions from the two neon isotopes (^{20}Ne and ^{22}Ne). Due to the high yield a tail function to the 7–6 line was added, with tail parameters fixed from our previous work [36]. A linear background model was adopted, since more complex forms did not yield a significant improvement in the reduced chi-squared value. Isotopic peaks corresponding to the same transition were constrained to improve fit stability, as detailed in the Methods section. Results for the principal KNe transitions are reported in Table I, with their statistical and systematic uncertainties. The value

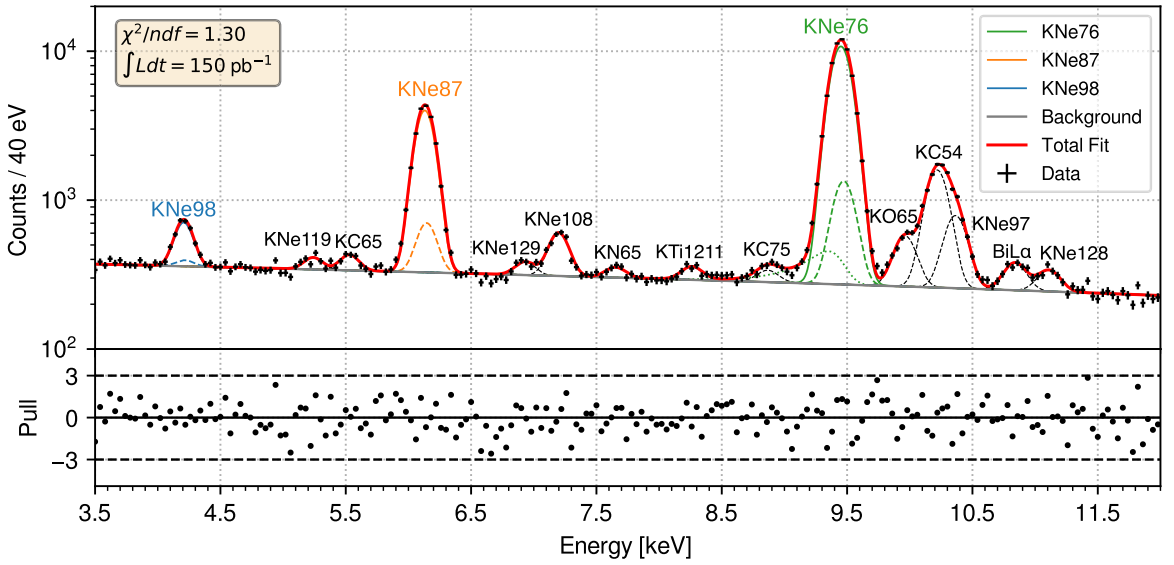


FIG. 3. Fit of the KNe x-ray spectrum (top panel) in the 3.5–12 keV range, showing the counts (black), the total fit (red), individual peak contributions (dashed black), and the background (gray). The main lines of KNe are highlighted: KNe76 (green), KNe87 (orange), and KNe98 (blue), each exhibiting a double component due to the presence of two different neon isotopes (^{20}Ne and ^{22}Ne). For the KNe76 peak, the tail contribution is also shown. The pull plot (bottom panel) displays the fit residuals normalized by the count errors.

for 6h-5g reported in [18] is also included in the table for completeness and as a reference for the calculations.

B. BSQED

We performed MCDFGME calculations to compare the experimentally measured transition energies with theoretical predictions. As summarized in Table I, the calculated transition energies, $E_{if}^{(\mathrm{calc.})}$, are presented alongside additional theoretical contributions. Specifically, we report QED contribution $E_{if}^{(\mathrm{QED})}$ to the transition energies, the sum of the first-order $E_{if}^{(\mathrm{QED1})}$ and second-order $E_{if}^{(\mathrm{QED2})}$ contributions [4].

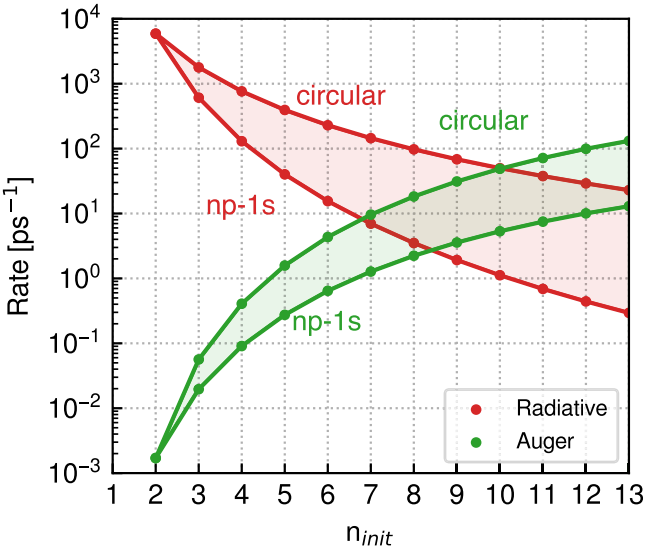


FIG. 4. Radiative and Auger transition rates for KNe as a function of the initial principal quantum number n. Rates for circular ($\ell = n - 1$) and $np \to 1s$ transitions are plotted as lines, while intermediate transitions fall in the corresponding shaded regions.

Accounting for vacuum polarization diagrams at all-order produces no noticeable effect on the transitions analyzed in this work. We also provide the isotopic shift for each transition, $\Delta E_{if}^{(\mathrm{isot.})}$, which was used to constrain the fitting procedure. For the most precisely measured line, the 7–6, we find $E_{76}^{(\mathrm{exp.})} = 9450.23 \pm 0.37\ (\mathrm{stat.}) \pm 1.50\ (\mathrm{syst.})\,\mathrm{eV}$, which is in excellent agreement with the calculated value of $E_{76}^{(\mathrm{calc.})} = 9450.28\,\mathrm{eV}$, where the QED contribution to the transition energy is 12.66 eV. The accuracy of the calculations depends solely on nuclear uncertainties, which have a negligible effect for these high-n transitions, as reported in the Supplemental Material [41], where results for different neon isotopes are reported in the Table S1 of the Supplemental Material [41]. We validate these results against the estimation of the energy shift $\Delta E_{if}^{(\mathrm{screen.})}$ due to the screening effect of an electron in the $1s$ orbital of neon. This contribution, always negative, reflects the reduced effective nuclear charge experienced by the kaon. The precise electron configuration of the atom at the moment of the x-ray transition is not known, since most electrons are expelled during the initial part of the cascade, first from the L shell, then from the K shell [44]. For all studied transitions, electron screening effects were found to be below 1 eV, with the 7–6 line shifted by only $-0.18\,\mathrm{eV}$. However, at the stage when radiative transitions become dominant ($n < 9$), KNe is expected to be highly ionized, as similarly indicated by the cascade calculations performed for KN [45]. This is further supported by the fact that, for these states, the wave functions are well contained within the Ne $1s$ electron orbital (see Fig. 5), indicating that the atom is expected to be highly ionized and therefore electron screening does not limit the comparison between the QED contribution and experiment. These findings establish KNe as a highly effective tool for probing BSQED, owing to its pronounced sensitivity to the magnitude of QED effects. To further establish the suitability of KNe, and kaonic atoms in general, for BSQED studies, we

TABLE I. Experimental transition energies $E_{if}^{(\mathrm{exp.})}$ for KNe obtained from the fit, including their statistical $\delta E_{if}^{(\mathrm{stat.})}$ and systematic $\delta E_{if}^{(\mathrm{sys.})}$ uncertainties, along with the calculated values $E_{if}^{(\mathrm{calc.})}$. The table also shows the QED contributions $E_{if}^{(\mathrm{QED})}$, isotopic $\Delta E_{if}^{(\mathrm{isot.})}$, and electron screening energy shifts $\Delta E_{if}^{(\mathrm{screen.})}$, not included in $E_{if}^{(\mathrm{calc.})}$), and uncertainty due to the PDG kaon mass $\Delta E_{if}^{(\mathrm{PDG})}$. All energies are given in eV.

Transition	$E_{if}^{(\mathrm{exp.})}$	$\delta E_{if}^{(\mathrm{stat.})}$	$\delta E_{if}^{(\mathrm{sys.})}$	$E_{if}^{(\mathrm{calc.})}$	$E_{if}^{(\mathrm{QED})}$	$E_{if}^{(\mathrm{QED1})}$	$E_{if}^{(\mathrm{QED2})}$	$\Delta E_{if}^{(\mathrm{isot.})}$	$\Delta E_{if}^{(\mathrm{screen.})}$	$\Delta E_{if}^{(\mathrm{PDG})}$
9l-8k	4206.97	3.43	2.00	4201.45	2.09	2.07	0.02	9.90	−0.38	0.11
8k-7i	6130.57	0.65	1.50	6130.31	5.09	5.05	0.04	14.45	−0.27	0.16
7i-6h	9450.23	0.37	1.50	9450.28	12.66	12.56	0.10	22.28	−0.18	0.24
6h-5g[a]	15673.30	0.52	9.00	15685.39	32.75	32.51	0.24	37.01	−0.11	0.40

[a]Reference [18].

performed a systematic set of calculations for multiple transitions in KNe up to $n = 15$. Figure 6 provides an overview of the QED contributions across a range of transitions, emphasizing both the sub-eV precision attainable with SDD detectors and the relevant energy window from 2 to 50 keV for the SIDDHARTA-2 experiment. The survey reveals a variety of QED contributions, with transitions such as 7–6 and 6–5 being particularly prominent. Notably, there are several additional transitions with $\Delta n = 1, 2, 3$ where transitions with second-order QED contributions of the order of the sub-eV precision are highlighted. Therefore, performing BSQED studies with kaonic atoms offers several advantages. For kaonic neon, the BSQED contribution to the 7i–6h transition amounts to 12.66 eV, corresponding to about 0.13% of the total transition energy, which is approximately one order of magnitude larger than in the muonic case [19]. As a result, BSQED effects in kaonic atoms can be resolved with higher statistical significance for a given experimental precision. In antiprotonic neon [9], although the larger mass further enhances QED contributions, the interpretation is limited by comparatively larger FNS effects, whose uncertainties reduce the achievable sensitivity to BSQED.

C. Kaon mass

We now investigate the impact of the 13-keV PDG kaon mass uncertainty on the size of the BSQED contributions on the KNe lines. We performed several MCDFGME calculations of the transition lines under examination, scaling for the value of the kaon mass. In Fig. 7 we vary the kaon mass by 50 keV with respect to the PDG value and report the variation of the transition energies. Only positive mass variations are shown, as we find the energy-mass relation to be linear at the ppm scale, while the BSQED contribution shows a dependence on the kaon mass more than two orders of magnitude smaller. Employing the 13-keV uncertainty from PDG, the theoretical uncertainties in transition energies are listed in the last column of Table I. Taken together, these results allow us to summarize the overall theoretical accuracy of the present calculations. For the high-n circular transitions considered here, finite nuclear size and recoil corrections are negligible, at the level of a few meV. Electronic screening effects, while not included in the calculated transition energies, were explicitly evaluated and conservatively treated as a systematic uncertainty below the eV level. The dominant theoretical uncertainty source arises from the present uncertainty on the charged kaon mass, which propagates to 0.24 eV for the 7–6 transition. This total theoretical uncertainty remains well below the experimental

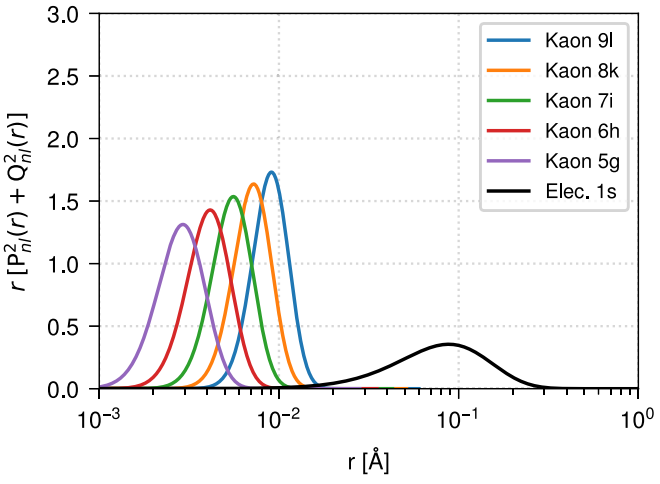


FIG. 5. Radial probability densities for selected kaonic neon states (9l, 8k, 7i, 6h, 5g) compared with the electron 1s wave function of neon. The kaonic wave functions are entirely contained within the electron 1s shell, demonstrating that the kaon orbits deep inside the electronic cloud.

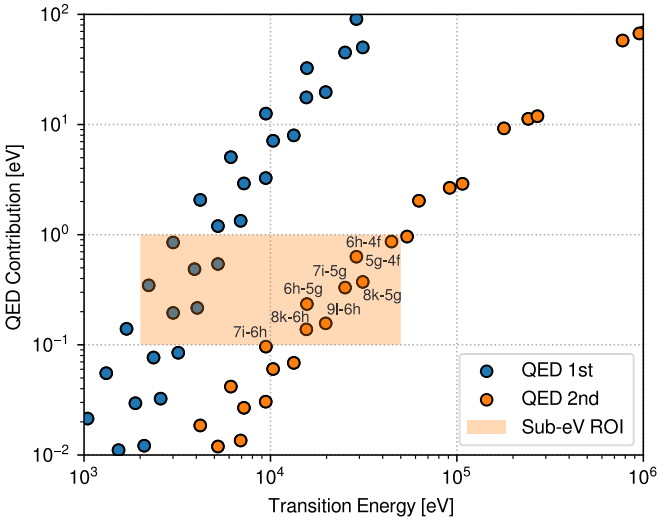


FIG. 6. First-order (blue) and second-order (orange) QED contributions as a function of transition energy for $K^{20}Ne$, all in eV. The sub-eV ROI highlights the region sensitive to the SDDs for transition energies in the 2–50 keV range and precisions in the 0.1–1 eV region.

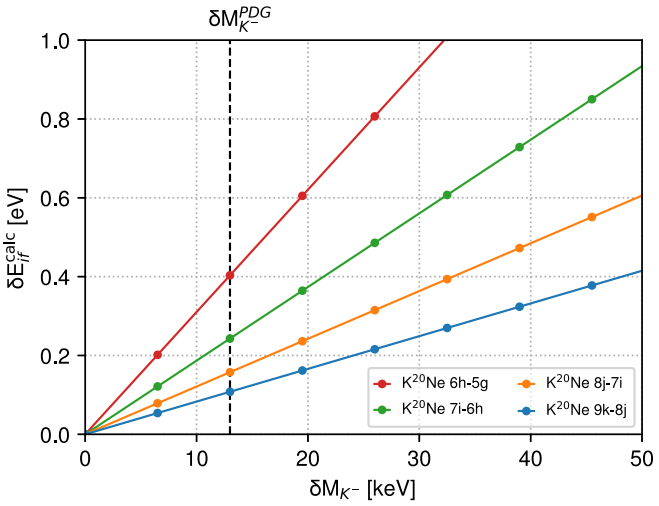


FIG. 7. Variation of the transition energies as a function of the kaon mass. The vertical dashed line indicates the current PDG uncertainty of 13 keV on the kaon mass.

systematic uncertainty and is more than an order of magnitude smaller than the corresponding BSQED contribution. Finally, we show the kaon mass uncertainty achievable with the sub-eV-precision 7–6 and 8–7 transitions of KNe. For each transition we applied the iterative procedure of Gall [27], with full propagation of both statistical and systematic uncertainties. Table II presents the resulting mass values and uncertainties for the 7–6 and 8–7 and their combination.

IV. CONCLUSIONS

In summary, the comparison of state-of-the-art MCDFGME calculations with recent SIDDHARTA-2 measurements at DAΦNE establishes kaonic atoms as a robust platform for testing BSQED in intermediate-Z systems. From the KNe x-ray spectrum fit, we extracted transition energies for the 7–6, 8–7, and 9–8 lines. The 7–6 and 8–7 transitions achieved sub-eV statistical precision, with the 7–6 line measured at 9450.23 ± 0.37 (stat.) $\pm$ 1.50 (syst.) eV, close to the MCDFGME prediction of 9450.28 eV, which includes a QED contribution of 12.66 eV. We also examined two potential systematic effects in the calculations: electron screening and the charged kaon mass uncertainty. For these transitions, the atom is expected to be highly ionized; however, electron screening from a residual 1s electron in neon introduces a shift of only about 1% of the QED contribution. Moreover, the current PDG uncertainty of 13 keV on the kaon mass induces only a few percent variation on transition energies. Together, these findings highlight that theoretical uncertainties remain at the percent level relative to the dominant QED contribution, making kaonic atoms well suited systems for precision BSQED tests in the intermediate-Z regime. In the end, we have shown that by combining the sub-eV-precision 7–6 and 8–7 transitions, it is possible to achieve a statistical uncertainty on the kaon mass below 20 keV, already approaching the precision of the two most accurate, but mutually inconsistent, measurements reported by the PDG. We note that the present experimental setup was optimized for the deuterium measurement; a dedicated campaign focused on kaonic neon could bring the total uncertainty on the kaon mass below 10 keV. This would be achievable by reducing systematic uncertainties through dedicated calibration of the KNe lines at 6 and 9 keV. Additionally, doubling the statistics and optimizing the kaon stopping efficiency specifically for a neon gas target would further lower the statistical uncertainty, potentially reaching the sub-10-keV level. Our results establish KNe as a benchmark for future BSQED tests and for improved determinations of fundamental hadronic properties. Further advances in detector technology and statistics, development of more refined cascade models, and the extension of this methodology to other intermediate-Z kaonic atoms promise even more stringent tests of QED in strong fields. Such progress will enable improved determinations of fundamental particle properties, including the charged kaon mass, and may shed light on discrepancies in precision spectroscopy, hadronic interactions, and the interplay between QED and the strong force in exotic atoms.

TABLE II. Kaon mass (M_{K^-}) extracted from different transitions, along with statistical ($\delta M_{K^-}^{\text{stat.}}$) and systematic ($\delta M_{K^-}^{\text{syst.}}$).

Transition	M_{K^-} [MeV]	$\delta M_{K^-}^{\text{stat.}}$ [keV]	$\delta M_{K^-}^{\text{syst.}}$ [keV]
7i-6h	493.674	19	78
8k-7i	493.699	52	121
7i-6h + 8k-7i	493.677	18	66

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DATA AVAILABILITY

The data that support the findings of this article are openly available [46], embargo periods may apply.

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X-Ray Emission from Atomic Systems Can Distinguish between Prevailing Dynamical Wave-Function Collapse Models

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In this work the spontaneous electromagnetic radiation from atomic systems, induced by dynamical wave-function collapse, is investigated in the x-ray domain. Strong departures are evidenced with respect to the simple cases considered until now in the literature, in which the emission is either perfectly coherent (protons in the same nuclei) or incoherent (electrons). In this low-energy regime the spontaneous radiation rate strongly depends on the atomic species under investigation and, for the first time, is found to depend on the specific collapse model.

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Introduction.—Quantum mechanics is our most successful physical theory, allowing us to understand and predict a large number of phenomena with extreme precision [1]. At the core of the theory lies the superposition principle, according to which systems are allowed to be in superposition of different states. The theory does not set any boundary on the limit of validity of this principle; yet, we do not observe superpositions of macroscopic objects. This well-known problem is exemplified by the Schrödinger's cat thought experiment [2].

It has been suggested that the linearity of quantum mechanics, from which the superposition principle directly follows, may break down at a certain scale [3–5]. This idea has been extensively developed in the framework of collapse models [6–9], which are phenomenological models that modify the Schrödinger equation by adding nonlinear and stochastic terms that naturally collapse the wave function in space. According to these models, microscopic systems are very weakly affected by the nonlinearities which, however, become dominant when atoms glue together to form larger and larger systems, this way solving the measurement problem. Among the several collapse models proposed, two are of particular relevance: the continuous spontaneous localization (CSL) model [10] and the so-called Diósi-Penrose (DP) model [11,12].

Spontaneous collapses must be random, in order to avoid the possibility of faster than light signaling [13]; this randomness manifests as a diffusive motion of the system [14], which corresponds to a random acceleration of the atoms and, hence, to the emission of radiation from their charged constituents. The experimental search of this spontaneous radiation was performed, for both CSL [15] and DP [16], in the energy domain of the γ rays, by comparing the measured radiation spectrum by high purity germanium crystals with the spontaneous emission rate predicted by the models for the atomic systems which constitute the experimental setup. The obtained strong bounds, combined with constraints provided by other experimental tests and theoretical considerations, are leading to a progressive falsification of the models in their Markovian formulation.

Theoretical efforts have being devoted to the development of non-Markovian collapse models [17–21], in order to counteract the runaway energy increase. These new models require the introduction of a cutoff frequency in the stochastic noise spectrum; for this reason, a systematic scan of the spontaneous radiation phenomenon, as a function of the decreasing energy, is mandatory.

The search for spontaneous radiation emitted by germanium crystals was performed in the x-ray domain in [22] [for $E \in (15\text{–}50)$ keV], and more recently in [23] [for $E \in (19\text{–}100)$ keV]. In [22] a formula for the expected spontaneous radiation rate from quasifree electrons was applied: the expected radiation depends on the energy as

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$1/E$ and is proportional to the number of quasifree electrons [24]. This formula is not suitable to describe the more complex phenomenology of the spontaneous radiation emitted by the whole atomic system; more refined calculations are presented in [15,16], where the CSL and the DP rates are calculated for an atomic system, in the limit in which the spontaneous photon wavelength λ_γ is between the nuclear dimension and the mean radius of the lower laying atomic orbit. Consistently, in [15,16] the data analyses are performed for photons energies in the range (1–3.8) MeV. The rate results to be proportional, for both CSL and DP, to $(N_p^2 + N_e)/E$, where N_p and N_e are, respectively, the number of protons and electrons of the atom under study. In this regime, different collapse models share the same expected shape for the energy distribution of the spontaneous emission rate, the scaling factor being proportional to combinations of constants of nature with the characteristic parameters of the models, which are λ and r_C (strength and correlation length of the collapse noise) for CSL and the correlation length R_0 for DP (the role of the strength being played by the gravitational constant G in this case). The latter theoretical rates were also assumed in the analysis [23], which set the strongest bounds on the parameters of the Markovian models, i.e., $\lambda/r_C^2 < (4.94 \pm 0.15) \times 10^{-1}\ \mathrm{s^{-1}\,m^{-2}}$, assuming that the white noise is coupled to the particle mass density, and $R_0 > (2.54 \pm 0.03) \times 10^{-10}$ m.

Given the importance of integrating the search of the spontaneous radiation signal in the high-energy domain of the γ rays to the one in the x rays, also in view of future experimental studies, scanning the cutoff energy of the non-Markovian models, we derive in this work the general expression of the radiation emission rates, for both Markovian and non-Markovian formulations of the CSL and DP models. We adopt a semiclassical approach which is valid above 1 keV (see, e.g., [17]), appropriate to the current experimental surveys; for lower energies a fully quantum mechanical analysis is required, which is under development. The general rates are found to exhibit a nontrivial energy dependence, which is strongly influenced by the interplay between the photons wavelengths, the radii of the electronic orbits, and the correlation length of the model under study. Since the correct general spectra strongly differ from the $1/E$ behavior in the x-ray range, a reanalysis of the data in Refs. [22,23] should be considered.

Interestingly, the spontaneous radiation energy spectrum is found, at the atomic λ_γ scale, to depend on the specific model of wave-function collapse under scrutiny. This finding opens new scenarios in the experimental investigation of the spontaneous radiation: a measurement sensitive to this signature of the collapse, would be able to recognize the most probable pattern of dynamical wave-function reduction.

CSL spontaneous emission rate, general expression.—The rate of the spontaneous radiation emitted by an atomic system, in the context of the Markovian CSL model, was derived in [15]:

$$\left.\frac{d\Gamma}{dE}\right|_t^{\mathrm{CSL}} = \frac{\hbar\lambda}{6\pi^2\epsilon_0 c^3 m_0^2 E}\sum_{i,j}\frac{q_i q_j}{m_i m_j} f_{ij}\,\frac{\sin(b_{ij})}{b_{ij}}, \quad (1)$$

where $b_{ij} = 2\pi|\mathbf{r}_i - \mathbf{r}_j|/\lambda_\gamma$, q_j, and m_j represent, respectively, the charge and the mass of the jth particle, at position $\mathbf{r}_j$. m_0 denotes the nucleon mass, ϵ_0 the vacuum permittivity, $\hbar$ and c are, as usual, the reduced Planck constant and the speed of light, E is the energy of the spontaneously emitted photon.

The term f_{ij} encodes the balance between the emitters' distances and the correlation length r_C. A generalized expression for f_{ij} is provided in Sec. A of supplemental material [25]:

$$f_{ij} = \frac{m_i m_j}{2r_C^2}\, e^{-\frac{(\mathbf{r}_i-\mathbf{r}_j)^2}{4r_C^2}}\left(3 - \frac{(\mathbf{r}_i-\mathbf{r}_j)^2}{2r_C^2}\right). \quad (2)$$

Using Eq. (2), and analyzing the contributions to the spontaneous radiation by the protons in the nucleus, the orbital electrons, and due to the combined electrons-protons emission—see Sec. B of supplemental material [25]—Eq. (1) turns to

$$\left.\frac{d\Gamma}{dE}\right|_t^{\mathrm{CSL}} = \frac{\hbar e^2\lambda}{12\pi^2\epsilon_0 c^3 m_0^2 r_C^2 E}\left\{3N_p^2 + 3N_e + 2\sum_{oo'\ \mathrm{pairs}} N_o N_{o'}\,\frac{\sin\left[\frac{\beta|\rho_o-\rho_{o'}|E}{\hbar c}\right]}{\left[\frac{\beta|\rho_o-\rho_{o'}|E}{\hbar c}\right]}\, e^{-\frac{\beta^2(\rho_o-\rho_{o'})^2}{4r_C^2}}\left(3 - \frac{\beta^2(\rho_o-\rho_{o'})^2}{2r_C^2}\right)\right.$$
$$\left. -2N_p\sum_o N_o\,\frac{\sin\left(\frac{\rho_o E}{\hbar c}\right)}{\left(\frac{\rho_o E}{\hbar c}\right)}\, e^{-\frac{\rho_o^2}{4r_C^2}}\left(3 - \frac{\rho_o^2}{2r_C^2}\right) + \sum_o N_o(N_o-1)e^{-\frac{(\alpha\rho_o)^2}{4r_C^2}}\,\frac{\sin\left(\frac{\alpha\rho_o E}{\hbar c}\right)}{\left(\frac{\alpha\rho_o E}{\hbar c}\right)}\left(3 - \frac{(\alpha\rho_o)^2}{2r_C^2}\right)\right\}, \quad (3)$$

where the electron pair distances are parametrized in terms of the mean radii of the atomic orbits ρ_o ($\rho_{o'}$) by means of the constants α and β which are given below. N_o represents the number of electrons in the oth orbit of the atom. Equation (3) represents a generalization of the spontaneous emission rate which was derived in Ref. [15]:

$$\left.\frac{d\Gamma}{dE}\right|_t^{\mathrm{CSL}} = \frac{\hbar e^2 \lambda}{4\pi^2 \epsilon_0 c^3 r_C^2 m_0^2 E}(N_p^2 + N_e), \tag{4}$$

under the condition that λ_γ is intermediate between the nuclear and atomic dimensions, i.e., the energy range under scrutiny belongs to the γ-ray domain. In that case, protons emit coherently (proportionally to the square of their number) and electrons emit independently (linear dependence). More complex is the situation described by Eq. (3).

If r_C exceeds the distance between the emitters (as confirmed by radiation experiments for a Markovian CSL [15,22]) then the stochastic field "shakes" them coherently. If λ_γ becomes also of the order of the mean orbit radii of the atom, then the electrons of the corresponding orbits start to emit coherently, i.e., quadratically. Nonetheless, the corresponding increase in the expected spontaneous emission rate is counteracted by the *cancellation*, among oppositely charged particles whose distance is smaller than λ_γ. In the limit in which λ_γ is also much bigger than the atomic size, Eq. (3) reduces to

$$\left.\frac{d\Gamma}{dE}\right|_t^{\mathrm{CSL}} = \frac{\hbar e^2 \lambda}{4\pi^2 \epsilon_0 c^3 m_0^2 r_C^2 E}[N_p^2 - 2N_p N_e + N_e^2], \tag{5}$$

which vanishes for neutral atoms.

Intermediate regimes for λ_γ and r_C, in comparison with $|\mathbf{r}_i - \mathbf{r}_j|$, give rise to a new, interesting pattern, in which the shape of the expected spontaneous radiation spectrum exhibits a nontrivial energy dependence, which is influenced by the atomic structure. This is exemplified in Fig. 1. The top panel of Fig. 1 shows (dashed line) the general spontaneous emission rate predicted by Eq. (3) compared to the simple (solid line) case described by Eq. (4), for germanium (Ge targets were indeed used in various experiments, e.g., [15,23]). The grey-shaded area corresponds to the values $\beta = 1.04$ and α spanning in the range (1–1.5), according to the literature [26,27], see Sec. B of supplemental material [25]. The same rates for a xenon target are shown in the bottom panel of Fig. 1. In Eq. (3) the value of r_C is set to 1.15×10^{-8} m, consistently with the results of Ref. [15], which are obtained by applying Eq. (4) in the γ-ray regime [(1–3.8) MeV], where Eq. (4) is an excellent approximation. $r_C = 1.15 \times 10^{-8}$ m corresponds to the intersection among the experimental bound and the theoretical constraint (corresponding, respectively, to the orange and gray lines in Fig. 4 of Ref. [15]). The mean radii of the orbits are obtained based on a density functional theory (DFT) [28] all-electron calculation, for an isolated atom; the DFT code GPAW [29] is adopted. The distributions are normalized to the common constant prefactors to evidence differences in shape. As expected, the simple (solid line) and the general (dashed line) rates converge for high energies (above 200 keV), where λ_γ becomes sizably smaller than the lower atomic orbit radii. Since r_C is much

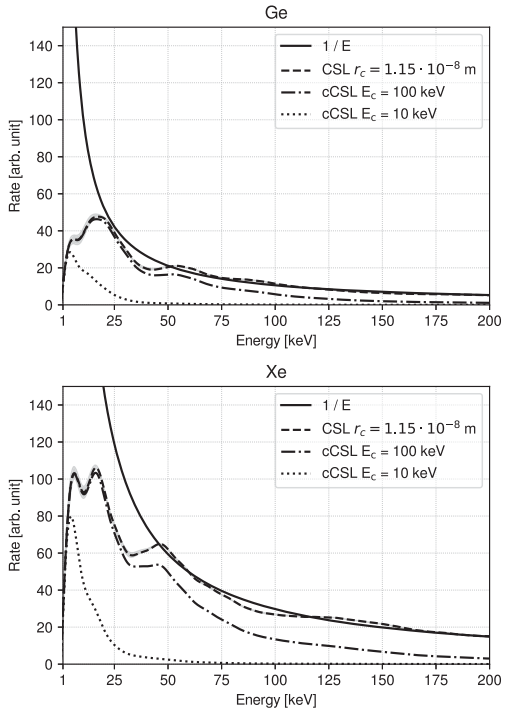


FIG. 1. The top panel of the figure shows (solid line) the $1/E$ dependence Eq. (4), for the spontaneous radiation rate of a Markovian CSL model, which is valid only in the high-energy domain. This is compared to the general rate in Eq. (3) (dashed line) for a prior value of the correlation length $r_C = 1.15 \times 10^{-8}$ m. The distributions are calculated for a germanium atom and normalized to the common constant prefactors. The bottom panel of the figure shows the shapes of the same rates, calculated for a xenon atom. The dotted and dash-dotted curves in the top and bottom panels, represent the corresponding spontaneous emission rates in Ge and Xe for a non-Markovian CSL model Eq. (6), when $E_c = 10$ keV (dotted) and $E_c = 100$ keV (dash-dotted).

greater than the size of the germanium atom, the x-ray regime is, instead, characterized by a balance among electrons and protons coherent emission and the cancellation of their contributions. The analyses performed in Refs. [22,23] should be reconsidered based on this low-energy complex pattern. On one side the smaller expected rate could result in less stringent bounds on the parameters of the CSL model, which could require collecting more statistics. On the other side, the sensitivity may be enhanced by the new structure of the energy spectrum of the spontaneous radiation, which could help to better disentangle the expected signal from the background components.

The general expression of the spontaneous emission rate in Eq. (3) encodes the phenomenology for future investigations of the spontaneous radiation at low energies (x rays). Comparison of the theoretical expectation with the measured spectra requires a recursive analysis: in the first step, a suitable prior has to be assumed for r_C, an updated value for r_C will be obtained, which will serve as input for

the new prior. The analysis should then be iterated till convergence of the r_C values, within the experimental sensitivity, is reached.

The dependence of the expected rate on the atomic structure becomes evident comparing the top panel of Fig. 1 with the bottom one, which shows as a dashed line the general spontaneous emission rate given by Eq. (3) for xenon (high sensitivity bounds on the spontaneous collapse could be set by the XENON experiment [30] by exploiting a xenon target); the solid line describes again the simple rate of Eq. (4). Equation (3) predicts a strong dependence of the spontaneous radiation yield on the atomic number Z; as such a survey of the spontaneous collapse induced emission over Z would greatly improve the experimental sensitivity on this, new physics, phenomenon.

The generalization of Eq. (3) to the non-Markovian case requires to multiply the right-hand side by the Fourier transform of the noise correlation function (see, e.g., [17,19–21,31] and the derivation of the colored DP model emission rate below). Assuming, e.g., an exponentially decaying noise correlation function $[f(t-s) = (\Omega/2)e^{-\Omega|t-s|}]$, characterized by a correlation time Ω^{-1}, the rate becomes

$$\left.\frac{d\Gamma}{dE}\right|_t^{\mathrm{cCSL}} = \left.\frac{d\Gamma}{dE}\right|_t^{\mathrm{CSL}} \times \frac{E_c^2}{E_c^2 + E^2}, \tag{6}$$

where $E_c = \hbar\Omega$ and cCSL denote results for a colored (non-Markovian) CSL model. The dotted and dash-dotted curves in Fig. 1 represent the general spontaneous emission rates for a non-Markovian CSL model Eq. (6), for $E_c =$ 10 keV (dotted) and $E_c = 100$ keV (dash-dotted).

DP spontaneous emission rate, general expression.— The rate of the spontaneous radiation emitted by an atomic system, according to a Markovian DP model, was derived in Ref. [16]:

$$\left.\frac{d\Gamma}{dE}\right|_t^{\mathrm{DP}} = \frac{\mathrm{Ge}^2}{12\pi^{5/2}\epsilon_0 c^3 R_0^3 E}(N_p^2 + N_e), \tag{7}$$

assuming a spontaneous photon wavelength which is much bigger than the nuclear size and much smaller than the lower lying atomic orbit mean radius. Note that Eq. (7) differs from the result presented in [16] by a factor 8π. This is because in [16] we adopted the convention introduced in [12] while we refer here to the original model introduced by Diósi [32], in which the factor 8π was not present.

The general structure of the rate is derived in Sec. C of supplemental material [25], where the non-Markovianity of the noise time correlation is also considered. For an exponential time correlation we have

$$\begin{aligned}\left.\frac{d\Gamma}{dE}\right|_t^{\mathrm{cDP}} &= \frac{G}{6\pi^2\epsilon_0 c^3 E}\sum_{i,j} q_i q_j f_{ij}\frac{\sin(b_{ij})}{b_{ij}}\frac{E_c^2}{E_c^2+E^2} \\ &= \left.\frac{d\Gamma}{dE}\right|_t^{\mathrm{DP}}\frac{E_c^2}{E_c^2+E^2},\end{aligned} \tag{8}$$

with G the Newton constant. The rate for the Markovian model is recovered in the limit $E_c \to \infty$. In analogy with the expected rate for the CSL model [Eq. (1)], the interplay between the particle mean distances and the wavelength of the spontaneously emitted photon is contained in the terms $\sin(b_{ij})/b_{ij}$. The dependence on the particle distances in relation to the correlation length of the model R_0 is instead specified by the terms f_{ij}. As it is shown in Sec. D of supplemental material [25], f_{ij} is a measure of the overlap between the mass densities of the particles ith and jth ($g_{i,j}$), whose spatial resolution is measured by R_0. In formula

$$f_{ij} = 4\pi\int d\boldsymbol{r} g_i(\boldsymbol{r}-\boldsymbol{r}_i, R_0) g_j(\boldsymbol{r}-\boldsymbol{r}_j, R_0). \tag{9}$$

Assuming Gaussian mass density profiles (following, e.g., [10,16] $g_i(\boldsymbol{r}-\boldsymbol{r}_i, R_0) = (2\pi R_0^2)^{-3/2} e^{-[(\boldsymbol{r}-\boldsymbol{r}_i)^2/2R_0^2]}$) and specifying the rate for the mean radii of the atomic orbits we obtain (see Sec. D of supplemental material [25]):

$$\begin{aligned}\left.\frac{d\Gamma}{dE}\right|_t^{\mathrm{DP}} = \frac{\mathrm{Ge}^2}{12\pi^{5/2}\epsilon_0 c^3 R_0^3 E}\Bigg\{ & N_p^2 + N_e + 2\sum_{oo' \text{ pairs}} N_o N_{o'} \frac{\sin\left[\frac{\beta|\rho_o-\rho_{o'}|E}{\hbar c}\right]}{\left[\frac{\beta|\rho_o-\rho_{o'}|E}{\hbar c}\right]} e^{\frac{\beta^2(\rho_o-\rho_{o'})^2}{4R_0^2}} \\ & + \sum_o N_o(N_o-1) e^{-\frac{(\alpha\rho_o)^2}{4R_0^2}} \frac{\sin\left(\frac{\alpha\rho_o E}{\hbar c}\right)}{\left(\frac{\alpha\rho_o E}{\hbar c}\right)} - 2N_p\sum_o N_o \frac{\sin\left(\frac{\rho_o E}{\hbar c}\right)}{\left(\frac{\rho_o E}{\hbar c}\right)} e^{-\frac{\rho_o^2}{4R_0^2}}\Bigg\}.\end{aligned} \tag{10}$$

Again, the simple rate described by Eq. (7) is a good approximation of Eq. (10) for energies belonging to the γ-ray domain. Figure 2 compares the general expression of the Markovian DP spontaneous emission rate [Eq. (10)] given by the dashed curve, with the simple expression in Eq. (7) given by the solid curve. The prior value $R_0 = 0.54$ Å is chosen, consistently with the result [16], which is obtained by applying Eq. (7) in the γ-ray regime

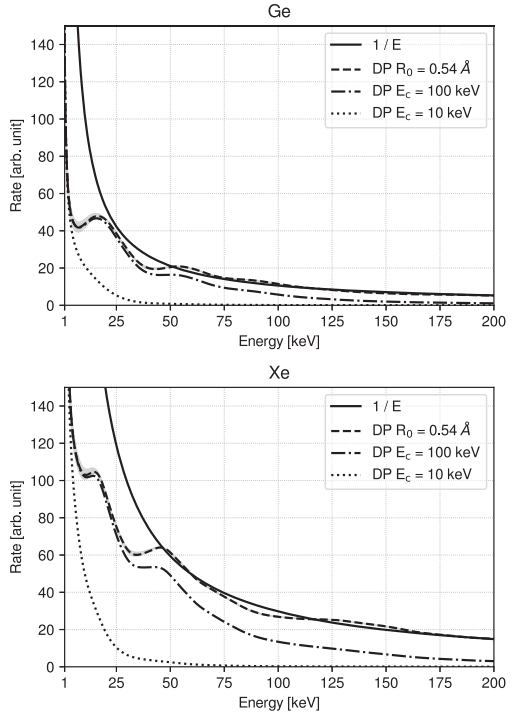


FIG. 2. Top panel of the figure shows (solid line) the $1/E$ dependence Eq. (7), for the spontaneous radiation rate of the Markovian DP model, which is valid only in the high-energy domain. This is compared to the general rate Eq. (10) (dashed line) for a prior value of the correlation length $R_0 = 0.54$ Å. The distributions are calculated for a germanium atom and normalized to the common constant prefactors. The bottom panel of the figure shows the shapes of the same rates, calculated for a xenon atom. The dotted and dash-dotted curves in the top and bottom panels, represent the corresponding spontaneous emission rates in Ge and Xe for a non-Markovian DP model, Eq. (8), when $E_c =$ 10 keV (dotted) and $E_c = 100$ keV (dash-dotted).

[(1–3.8) MeV], where Eq. (4) is an excellent approximation. A strong departure from the approximate expression is evident, at low energy, for both germanium (top panel) and xenon (bottom panel) atoms. A significant dependence of the generalized rate on the atomic number is evident for the DP model as well. The dotted and dash-dotted curves in Fig. 2 represent the general spontaneous emission rates for a non-Markovian DP model Eq. (8), for $E_c = 10$ keV (dotted) and $E_c = 100$ keV (dash-dotted).

Discussion.—Figures 1 and 2 summarize our findings and unveil the most interesting consequence of the cancellation phenomenon, for which the contribution to the spontaneous radiation emission from oppositely charged particles, whose distance is exceeded by the model's correlation length and by the observed photon wavelength, cancels. In the low-energy regime, for correlation lengths of the models of the order, or bigger, than the atomic orbits radii, the shapes predicted for the spontaneous emission rates distributions of the CSL and DP models strongly differ. This is due to both the different mathematical structure of the terms f_{ij} and the different values of the correlation lengths, r_C and R_0, of the two models. In the simple scenario in which λ_γ is much smaller than the atomic size, any difference is washed out and the shapes of the spontaneous radiation rates of the two models just differ by a scaling factor.

The energy ranges analyzed in Refs. [22,23] do not allow to unveil differences among the spontaneous emission spectra of the CSL and the DP. Recent experimental works [33,34] present data in the ranges $E \in (5\text{–}30)$ keV (with lower statistics with respect to Ref. [23]) and $E \in (1\text{–}140)$, respectively. The analysis of these data, based on the predicted rates Eqs. (3) and (10), would provide a first test of this prediction for a Ge target [33] and a Xe target [34]. A measurement complementary to [33], with improved statistics, and including the energy interval $E \in (1\text{–}5)$ keV, would be important. Moreover, the prediction of a strong dependence of the spontaneous emission rate on the atomic structure should be exploited by performing a dedicated experiment, performing a search for spontaneous radiation signal by scanning over several targets (under the same experimental conditions) in the energy ranges in which the CSL and DP rates differ for the considered targets. Such a measurement would further improve the sensitivity and would allow us to distinguish an eventual signal detection from systematic effects.

In this Letter, we derived the fundamental theoretical formulas for these future investigations, both for the white noise models and their non-Markovian generalizations.

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